CSCE 3701 · TEAM CLOSET · MILESTONE 2 FINAL

Software Requirements Specification

Version 1.3 · revision 5 · application build v1.6

Team	Habiba · Salma · Bashar · Karim · Youssef
Date	10/29/2026
How to read it	The first pages (header “v1.3 (rev. 5)”) are revision 5. The v1.2 revision 4 document follows unchanged as the tracked baseline; where revision 5 supersedes it, R5.1 says so — the newer statement wins.

Revision history (revision 5 added)

Author	Date	Change	Version
Team (final pass)	10/28/2026	Revision 4 (build v1.5): proactive intelligence FR-81 – FR-101	1.2
Team (corrections)	**10/29/2026**	**Revision 5 (build v1.6): one garment lifecycle, Care (repairs, alterations, tailor), Borrow & lend, donate-only Donation, Home Wardrobe status, UI entry points, purchase check by name, Ask Closet 35 functions. FR-102 – FR-113, changes C-85 – C-97.**	**1.3**

R5.1 What changed in revision 5

Baseline statement (v1.2)	Now (v1.3)
FR-27 – FR-29: five statuses (Available, Laundry, Borrowed, Packed, Unavailable)	FR-102: nine states, 21 transitions; Donated is an archive, not a state
FR-83: Ready to donate / give / sell as a separate rehoming record	FR-106: donate only; Ready to donate is a state; Donation area
FR-90: borrow tracker on the piece page and Home	FR-105: Borrow & lend area with a lend form and history
“Repair” = status Unavailable with a free-text note	FR-103, FR-104: Care → Repairs & alterations, tailor, reminder, Mark collected
FR-86 – FR-88, FR-93, FR-96 reachable only through Ask Closet or the API	FR-109, FR-110, FR-113: visible entry points in the app
FR-94: purchase check by category and colour	FR-111: also by item name; five named result blocks
FR-99: 31 functions	FR-112: 35 functions
SDD v1.2 design decision “rehoming is not a status”	Reversed (SDD v1.3 R5.2)

R5.2 Section 4.22 · Garment lifecycle, care, lending and donation

Release: **Sprint 3 · built early in v1.6** (labelled “Sprint 3 · preview” in the app). Stories US-112 – US-117 are **proposed, not yet in Jira**.

ID	Requirement	Story	Screen
FR-102	Every piece shall be in exactly one of nine lifecycle states — Available, Laundry, Borrowed, Packed, Needs repair, Needs alteration, At tailor, Ready to donate, Unavailable — and move only along the 21 allowed transitions. Only Available is wearable. Donated is not a state: the piece leaves the active closet (archived with reason Donated, wear history kept).	US-112	Every screen (status badge, Closet filter)
FR-103	The user shall mark a piece Needs repair or Needs alteration with a reason (≤ 80 chars) and optional notes, see it under Care → Repairs & alterations, and close it with Fixed at home (→ Available) or Donate (→ Ready to donate).	US-113	Piece → More actions → Repair / Alter · Care → Repairs & alterations
FR-104	The user shall send a piece that needs care to a tailor with tailor name, date sent, expected return date and optional cost. While At tailor it is left out of outfits, plans and trips. On the expected day the Notification Center shows TAILOR_READY with Mark collected and Remind me later (1–30 days). Mark collected sets Available and keeps the care history.	US-113	Care → Repairs & alterations (At tailor) · Home · Notifications
FR-105	The Borrow & lend area shall list pieces currently lent (person, date lent, due date, note, days out, overdue first) and the returned history. Lend item records person (required), date lent and due date; Mark returned sets Available and closes the loan. A lent piece is excluded from outfits and raises a Planner conflict.	US-114	Wardrobe → Borrow & lend · Piece → More actions → Lend item
FR-106	Donation is the only way to pass a piece on (no give or sell). Donate sets Ready to donate (still owned, left out of outfits, duplicates, gaps and purchase impact); the Donation area lists Ready to donate and the Donated history; Mark as donated archives the piece with reason Donated and adds no replacement to the Shopping bag; Cancel donation sets Available. Analytics shows the donated count.	US-115	Wardrobe → Donation · Still love it? · Piece → More actions → Donate · Analytics
FR-107	Home shall show a Wardrobe status summary: total pieces, ready to wear, and one clickable line per non-empty state (laundry, borrowed, packed, needs repair / alteration, at tailor, ready to donate) that opens the place where that state is managed.	US-112	Home → Wardrobe status
FR-108	Availability conflict alerts shall cover every non-wearable state and name it (“packed”, “lent out”, “at the tailor”, “waiting for an alteration” …) with the expected date written as a day (“Mon 2 Nov”) and the correct verb for plural garments.	US-100 (changed)	Home “Closet noticed” · Planner · Notifications
FR-109	The repeat insight (FR-87) shall be visible on the outfit card (Keep it · Wear again · Change one piece · Show an alternative); Outfit history shall offer memory search, Wear again and Create a variation (FR-88).	US-98, US-99 (UI)	Style me outfit card · Analytics → Outfit history
FR-110	Style me shall offer an “I have nothing to wear” rescue for an occasion and date: three Available looks for its dress code and weather; a gap is named only when no complete look exists (FR-96).	US-107 (UI)	Style me → Rescue panel · Ask Closet

ID	Requirement	Story	Screen
FR-111	The purchase check shall use the item name (≤ 40 chars) as well as category and colour, and count only owned pieces (not Ready to donate). The simulation shall show five blocks: New outfit combinations · Similar items owned · Wardrobe gap · Main occasions · Recommendation; nothing is saved.	US-105 (changed)	Shopping → Do I need this? What would it add?
FR-112	Ask Closet shall map each message to one of 35 whitelisted functions, adding care_list (“What needs repair?”, “What is at the tailor?”), lend_piece (“Lend my black blazer to Nour until Friday”), donation_list, donate_piece, donate_done, unpacked (“What haven’t I packed?”), and tailor answers in availability (“When is my black blazer back from the tailor?”). Questions read state only.	US-110 (changed)	Ask Closet
FR-113	The Planner shall offer Plan several days (FR-86) and the Closet gaps view shall show each gap’s confidence (HIGH / MEDIUM / LOW) and reasons (FR-93).	US-97, US-104 (UI)	Planner → Plan several days · Closet → Closet gaps

R5.3 Non-functional and business rules

ID	Requirement	Verified by
NFR-17 (extended)	One source of truth for the lifecycle: the states, transitions, labels and the questions “wearable / owned / leaving / in care” live only in shared/lifecycle.js; every status change (including lending, care and donation) goes through Availability.setStatus, which keeps loans and care jobs in step.	ST-01 – ST-06 · LF-01 – LF-04 · LC-01 – LC-12 · code review
NFR-18 (extended)	Offline parity covers Care, Borrow & lend and Donation: the offline demo runs the same services and schemas; the lifecycle journey passes unchanged against it.	OF-10 · OF-11 · e2e/lifecycle_journey.py (offline)
NFR-19	One calendar day for the app: “today” is the UTC day on the server, in the seed and in the E2E journeys, so dates such as lent on, sent on and expected back never disagree across midnight.	LC-10 · defect D-18

ID	Rule
BR-12 (changed)	A donated piece never creates a replacement on the shopping list. Give away and sell are no longer offered (v1.5 rows with those reasons are still readable).
BR-14	A piece that needs care, is at the tailor, is lent out or is ready to donate is never recommended, planned or packed until it is Available again.

R5.4 State model

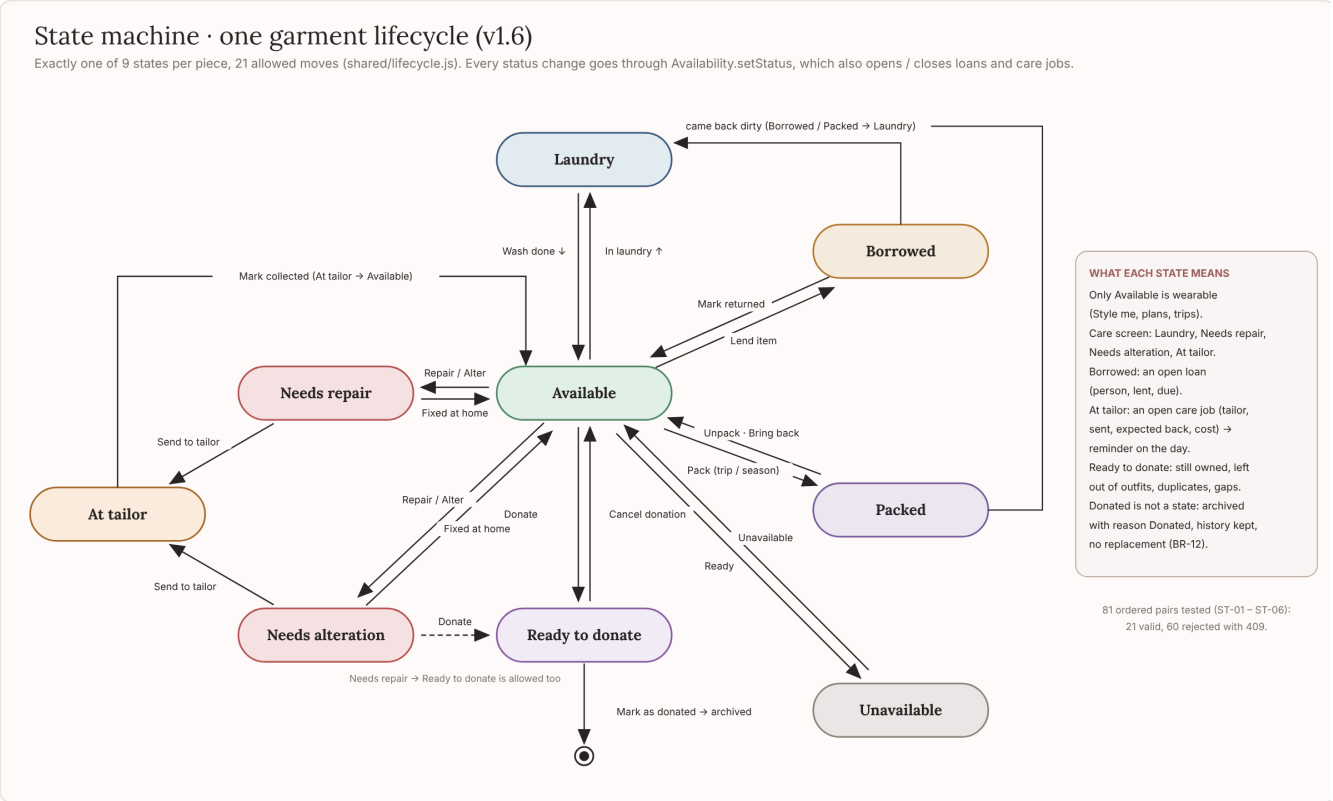


Figure R5-1 (D24). The one garment lifecycle (supersedes D21 and the v1.1 state machine figure).

R5.5 Change log C-85 – C-97

ID	Type	Section	Change
C-85	Added	4.22	One garment lifecycle with nine states and 21 transitions (FR-102); supersedes the five-status machine of FR-27 – FR-29 and the separate rehoming intention.
C-86	Added	4.22	Care: Repairs & alterations and the tailor flow with return reminder (FR-103, FR-104).
C-87	Changed	4.21 FR-90	Borrow tracker → Borrow & lend area with lend form and history (FR-105).
C-88	Changed	4.21 FR-83 · BR-12	Donate / give / sell → donate only, Ready to donate as a state, Donation area (FR-106).
C-89	Added	4.22	Home Wardrobe status with clickable counts (FR-107).
C-90	Changed	4.21 FR-89	Conflict alerts name every non-wearable state with a readable date (FR-108).
C-91	Changed	4.21 FR-86 – FR-88, FR-93, FR-96	UI entry points for multi-day plans, repeat insight, outfit memory, gap confidence and event rescue (FR-109, FR-110, FR-113).
C-92	Changed	4.21 FR-94	Purchase check uses the item name; simulation shows five named blocks (FR-111).

ID	Type	Section	Change
C-93	Changed	4.21 FR-99	Ask Closet 31 → 35 functions (FR-112).
C-94	Added	5.4	NFR-17 and NFR-18 extended; NFR-19 one calendar day.
C-95	Added	5.5	BR-12 changed; BR-14.
C-96	Added	6.3	Data: migration 007 (item_status CHECK 9 states, availability.care_jobs, availability.loans; wardrobe.rehoming converted to Ready to donate, then dropped).
C-97	Changed	6.4	Traceability extended to FR-102 – FR-113 (Closet_Traceability_Matrix_v1.6.xlsx).

R5.6 Traceability (excerpt)

FR	API	Tests	Demo step
FR-102	PATCH /items/:id/status · GET /vocabulary	ST-01 – ST-06 · LF-01 – LF-04 · API-S01 – S06	all
FR-103	POST /items/:id/care · POST /items/:id/care/done · GET /care	LC-01 · LC-04 · LF-05 · OF-10 · E2E I03	3
FR-104	POST /items/:id/care/tailor · GET /intelligence/tailor · POST /intelligence/tailor/:itemId/remind · POST /items/:id/care/done	LC-02 · LC-03 · LC-04 · ST-06 · E2E I03 – I05	3
FR-105	POST /items/:id/lend · POST /items/:id/return · GET /lending	LC-10 – LC-12 · API-PR40 · OF-11 · E2E I06 – I07	4
FR-106	POST/DELETE /items/:id/donate · POST /items/:id/donate/done · GET /donation	API-PR02 · API-PR03 · API-PR92 · LF-04 · LF-12 · E2E s08 – s09, I08 – I09	7
FR-107	GET /items	E2E I01 · stateful status check	1, 13
FR-108	GET /intelligence/monitor · GET /notifications	API-K01 – K04 · API-PR30 · API-PR32 · LC-03 · LC-11 · LF-06	10
FR-109	POST /intelligence/repeat · GET /intelligence/memory	API-PR60 · API-PR62 · IN-30 – IN-34	11
FR-110	POST /intelligence/rescue · POST /chat (event_rescue)	LC-23 · LC-30 · LF-10	—
FR-111	POST /shopping/check · POST /shopping/impact	API-PR71 · IN-62 · LC-22 · E2E lifecycle step 6	9
FR-112	POST /chat · POST /chat/undo/:id	LF-10 (22 prompts) · LF-11 · LF-12 · API-PR90 – PR93 · E2E s10	12
FR-113	POST /intelligence/plan-days · GET /intelligence/gaps	API-PR21 · API-PR72 · LC-20	10

Full matrix: `Closet_Traceability_Matrix_v1.6.xlsx`. Final run: 599/599 tests passing (how they were executed: Test Plan and Report v1.3).

> The baseline v1.2 revision 4 specification follows.

CSCE 3701 SOFTWARE ENGINEERING · THE AMERICAN UNIVERSITY IN CAIRO

Software Requirements Specification for Closet

Version 1.2 · revision 4 (Milestone 2 final) · application build v1.5

Prepared by	Habiba Khashaba, Salma, Bashar, Karim and Youssef (Team Closet)
Instructor	Dr. Sherif Aly
Date	10/28/2026
How to read it	The first pages (header “v1.2 (rev. 4)”) are revision 4: what changed, the new section 4.21 and the change log entries C-70 – C-84. Then the v1.1 revision 3 specification follows unchanged as the tracked baseline (header “Version 1.1 (rev. 3)”, its own page numbers). Where revision 4 supersedes a v1.1 statement, the table in R4.1 says so — the newer statement wins.

Revision history (revision 4 added)

Name	Date	Reason for changes	Version
Habiba Khashaba (team)	Sep 2026	Milestone 0: problem, users, first feature list	0.1
Habiba Khashaba (team)	10/13/2026	Milestone 1 baseline	1.0
Habiba Khashaba (team)	10/25/2026	Milestone 2: separate SRS (IEEE / Wiegers), Admin actor, authentication and authorization requirements, US-61 to Sprint 2	1.1
Habiba Khashaba (team)	10/26/2026	Revision 2 and 3: core flow first, deletion reasons, shopping list, try-on on the user's photo, Suggest a fix (C-50 – C-68)	1.1 rev. 2–3
Team (final pass)	**10/28/2026**	**Revision 4 (build v1.5): proactive wardrobe intelligence FR-81 – FR-99, three genuinely different looks FR-100, one record per saved look FR-101, Ask Closet with 31 functions, NFR-16 – NFR-18, BR-12 – BR-13. Changes C-70 – C-84.**	**1.2**

> **How changes are tracked.** Revision 4 does not rewrite the baseline. Every added or changed requirement is listed here with its ID; the change log addendum (R4.6) continues Appendix D. In the baseline pages, green text = added after v1.0 and red strike-through = removed (unchanged convention).

R4.1 What changed in revision 4

ID	Type	Where	Change
C-70	Added	4.21	Proactive wardrobe intelligence: FR-81 – FR-99 (built early as Sprint 3 previews in v1.5).
C-71	Changed	4.7 FR-78 / US-19	"no two share more than 2 pieces" → FR-100: at most one shared MAJOR garment; diversity-aware ranking.
C-72	Changed	4.12 FR-43 / US-26	Saved look identity (FR-101): Save + Favourite never create two records.
C-73	Changed	4.14 FR-51	"13 14 whitelisted functions" → 31; questions read state; out-of-scope answer (FR-99).
C-74	Changed	4.16 FR-60	Purchase analysis counts every owned piece (laundry included); distinct outfits only (FR-94).
C-75	Changed	4.3 FR-15 · FR-71	Delete reasons: + Donated, Given away, Sold / rehomed — only through FR-83, never a replacement (BR-12).
C-76	Added	5.4	NFR-16 determinism, NFR-17 one source of truth for status, NFR-18 offline fallback parity.
C-77	Added	5.5	BR-12, BR-13.
C-78	Changed	4.18 FR-64 – FR-66	Packing Mode → trip capsule with luggage limit, reuse and remove/re-plan (FR-95).
C-79	Changed	4.17 FR-62 – FR-63	Notification Center: types, priorities, dedupe keys, snoozes, ≤ 2 on Home (FR-98).
C-80	Added	3.3	GET /vocabulary now also returns STATUS_TRANSITIONS; the web app no longer has its own copy.
C-81	Added	6.3	Data: migration 006 (price, currency, purchase_date, wardrobe.rehoming, intelligence.snoozes, wears.event_title / note, events.planned_temp_c / planned_condition, profiles.allow_repeats).
C-82	Added	Appendix B	Use-case diagram D19 (proactive intelligence) added; D1 (with Admin) unchanged.
C-83	Changed	6.4	Traceability extended to FR-81 – FR-101 (Closet_Traceability_Matrix_v1.5.xlsx).

ID	Type	Where	Change
C-84	Changed	AS-3	Without an API key the keyword parser answers the same 31 functions.

Superseded statements in the baseline pages

Baseline statement (v1.1 rev 3)	Now (v1.2 rev 4)
US-19 acceptance: “no two share more than 2 pieces”	FR-100: any two looks share at most **one major garment** when the closet allows
FR-51 / AS-3: “13 14 whitelisted functions”	FR-99: **31** whitelisted functions
FR-15: six deletion reasons for a user delete	unchanged for a user delete (4 reasons in the app); Donated / Given away / Sold / rehomed only through FR-83
FR-64: one outfit per trip day	FR-95: trip capsule with luggage limit, reuse limits and remove / re-plan
FR-60 / US-88: purchase check uses Available pieces	FR-94: every owned piece counts; distinct outfits only

R4.2 New section 4.21 · Proactive wardrobe intelligence

4.21.1 Description and priority. Closet evolves from “tell me what to wear” to “I understand your wardrobe and warn you before something becomes a problem”. The features reuse the existing data — item status, wear history, weather, planner, recommendations, analytics and shopping — through one shared rule layer (*Wardrobe Insights*, SDD 5.14). Priority as agreed with the product vision: core intelligence (monitor, duplicates, gap confidence, memory) High; laundry, packing, event rescue and donation Medium; seasonal rotation, borrow tracker, repeats and the health explanation Low. Release: **Sprint 3 · built early in v1.5** (labelled “Sprint 3 · preview” in the app). The proposed stories US-94 – US-111 are **not yet in Jira** (Proposed_Future_Backlog.md).

4.21.2 Stimulus / response.

- Home → “Still love it?” → Style it · Keep it · Donate · Not now → three looks / snooze / donate flow.
- Planner and Home → “It’s now expected to be 30 °C instead of 24 °C — your navy blazer may be too warm” → Adjust outfit → only the blazer changes.
- Ask Closet → “Do I need to do laundry before Monday?”, “Pack me for 5 days in Paris”, “What did I wear to Salma’s birthday?” → answers from real data.
- Add a piece → Review → “You already own 2 similar white tops” → Add anyway · View similar · Cancel.

4.21.3 Functional requirements.

ID	Requirement	Story	Release
FR-81	The system shall identify long-unworn pieces: Available, not being rehomed, added 30+ days ago, not snoozed, in season (weather + calendar), and unworn for 90 days (everyday) or 180 days (special-occasion pieces, dress codes ≥ 4 only). A never-worn piece counts from the day it was added.	US-94	S3 (built early, v1.5)
FR-82	For a long-unworn piece the system shall offer Style it (three looks that all contain it), Keep it (no prompt for 90 days, optional reason), Donate (opens FR-83) and Not now (14 days). Nothing changes without the user’s choice.	US-94	S3 (built early, v1.5)
FR-83	The system shall let the user mark a piece Ready to donate / give / sell (still owned, left out of new outfits), cancel it, or confirm it has left; confirming removes it from the active closet with reason Donated, Given away or Sold / rehomed, keeps the history row, and never adds a replacement to the shopping list.	US-95	S3 (built early, v1.5)
FR-84	The system shall suggest seasonal rotation from the temperature the user will feel (≥ 27 °C summer, ≤ 17 °C winter) and each piece’s season: store opposite-season Available pieces not worn in 21 days; bring back pieces in seasonal storage. Pack for season sets status Packed (note “Seasonal storage”); Bring back sets Available; Not now hides a piece for 30 days. Never automatic; at most one seasonal notice on Home.	US-96	S3 (built early, v1.5)
FR-85	The system shall warn when, for an event in the next 7 days, at least two of the user’s three most-worn tops or bottoms that suit the event’s dress code are in the laundry; Wash done removes the warning.	US-97	S3 (built early, v1.5)
FR-86	The system shall plan outfits for several days without reusing a top, bottom or dress before a wash (shoes and accessories may repeat), unless the user allows repeats; when the closet runs out it repeats as little as possible and names the repeated piece.	US-97	S3 (built early, v1.5)
FR-87	The system shall recognise a near-identical look (≥ 3 shared main pieces) worn in the last 7 days and offer Keep it / Change one piece / Show an alternative, unless the user allows repeats; a look rated ≥ 4★ is offered positively as a repeat.	US-98	S3 (built early, v1.5)
FR-88	The system shall remember the context of a worn outfit (event title, occasion, date, temperature, condition, rating, optional note), find it from natural words (“Salma’s birthday”, “last Friday”, “liked most”), and use highly rated looks for the same occasion at ± 4 °C as one ranking signal capped at +3 per piece.	US-99	S3 (built early, v1.5)
FR-89	The system shall monitor planned outfits in ONE place for (a) a piece whose CURRENT status is not Available and (b) a material forecast change since planning (|Δ| ≥ 5 °C, or rain appearing); it shall offer to change only the affected piece using Change One Piece, store the new forecast with the plan, and never alert for smaller changes.	US-100	S3 (built early, v1.5)
FR-90	The system shall show who has a borrowed piece, since when and when it is expected back; overdue pieces raise one reminder with Mark returned and Remind me later (3 days); Mark returned sets Available.	US-101	S3 (built early, v1.5)
FR-91	The system shall explain the Wardrobe Health score: the parts that raise it and the three biggest that lower it, each with its points and data; the same wardrobe always gives the same score.	US-102	S3 (built early, v1.5)
FR-92	Before a piece is added the system shall check structured metadata (category, colour, garment type, style, dress code, season) for similar owned pieces and show “You already own …” with Add anyway, View similar and Cancel.	US-103	S3 (built early, v1.5)
FR-93	The system shall give every wardrobe gap a confidence (HIGH, MEDIUM, LOW), a priority (Essential, Useful, Optional) and its reasons (missing pieces, pieces waiting for it, upcoming occasions, stand-ins one level away); a gap is never HIGH when two or more equivalents exist one level away.	US-104	S3 (built early, v1.5)
FR-94	The system shall simulate a purchase without saving it: new distinct outfit combinations with what the user owns (all owned pieces count, including the laundry; pieces being rehomed do not), compatible pieces, occasions, similar pieces owned, gap filled and a recommendation (Useful purchase, Optional, Probably unnecessary, Low wardrobe impact).	US-105	S3 (built early, v1.5)

ID	Requirement	Story	Release
FR-95	The system shall build a trip capsule: one outfit per day preferring pieces already packed; carry-on limit 8 clothing pieces and 2 pairs of shoes; a top, bottom or dress worn at most twice (three times with laundry at the destination); remove a piece and re-plan; Mark as packed changes each REAL piece to Packed (no copies).	US-106	S3 (built early, v1.5)
FR-96	For an urgent event (“I have a wedding tomorrow”) the system shall first build the three strongest Available looks for its dress code and weather; only if none is possible shall it name the real missing category and link to Shopping.	US-107	S3 (built early, v1.5)
FR-97	The system shall accept an optional purchase price, currency and date, and show cost per wear = price ÷ confirmed wears (“Wear this” only — not saved, planned or recommended looks); with no wears it shall say so, never divide by zero; best value requires ≥ 3 wears.	US-108	S3 (built early, v1.5)
FR-98	The Notification Center shall hold every proactive notice in one list with a type (WEATHER_CHANGE, AVAILABILITY_CONFLICT, LAUNDRY_WARNING, BORROW_RETURN, SEASONAL_ROTATION, FORGOTTEN_DONATION …), priority (HIGH / MEDIUM / LOW), related item or event, date, read state and primary action; stable keys prevent duplicates, a donation notice appears at most once a week, and Home shows at most two notices.	US-109	S3 (built early, v1.5)
FR-99	Ask Closet shall map each message to exactly one of 31 whitelisted functions (outfits, change a piece, status, availability questions, laundry, borrowed, donate / keep, seasonal, memory, repeats, health, monitor, event rescue, trip, value, shopping …), resolve “it / this / that” from the conversation, read state without changing it for questions, and answer out-of-scope messages with “I can help with your wardrobe, outfits, laundry, shopping, weather and planner.”	US-110	S3 (built early, v1.5)
FR-100	The three recommended looks shall be distinct styling directions: when the closet allows, any two share at most one major garment (top, dress, bottom, layer, shoes); accessories may repeat; a style-this-piece set keeps the chosen piece in all three and is otherwise diverse; quality stays within one failed criterion of the best look.	US-19 (changed)	S2 (changed)
FR-101	A saved look shall be identified by its user and its set of pieces: Save then Favourite, or Favourite then Save, reach the same record; Save never removes a favourite; unfavourite keeps it under All looks.	US-26 (changed)	S2 (changed)

R4.3 Non-functional requirements and business rules added

ID	Requirement	Verification
NFR-16	Determinism of the demo: from a fresh seed with WEATHER_MODE=estimate the same looks, tray, forgotten-piece looks, Mix & Match scores and Ask Closet functions are produced on every run.	npm run demo (demo:check) on 3 fresh seeds
NFR-17	One source of truth: an item's status is stored only by Availability; every screen, the planner monitor, the engine and the web app's allowed moves (GET /vocabulary) read it — no copies that can go stale.	API-PR30 · E2E s02–s03 · code review
NFR-18	Offline fallback parity: the offline demo runs the same services and rules on in-memory repositories; the stateful browser journey passes unchanged against it.	OF-01 – OF-09 · e2e/stateful_journey.py (offline)
NFR-05 (extended)	Privacy: purchase price is optional personal data, never required, editable and removable; donation is never connected to a third party.	API-PR50 · code review

ID	Business rule
BR-12	A piece that the user donates, gives away or sells never creates a replacement on the shopping list (like "No longer my style").
BR-13	No proactive feature changes state on its own: packing for the season, donating, keeping, adjusting a planned outfit and learned preferences all wait for the user.

Security requirements — unchanged, now tested on every new route

NFR-04 applies to every v1.5 route: authentication (JWT), validation (schemas, 400), ownership (another user's piece, plan or trip → 404), role checks. Examples: API-PR01 (keep someone else's piece → 404), API-PR03 (rehome someone else's piece → 404, invalid method → 400), API-PR50 (negative price → 400), OF-02 (offline demo without the session → 401).

R4.4 Use cases added

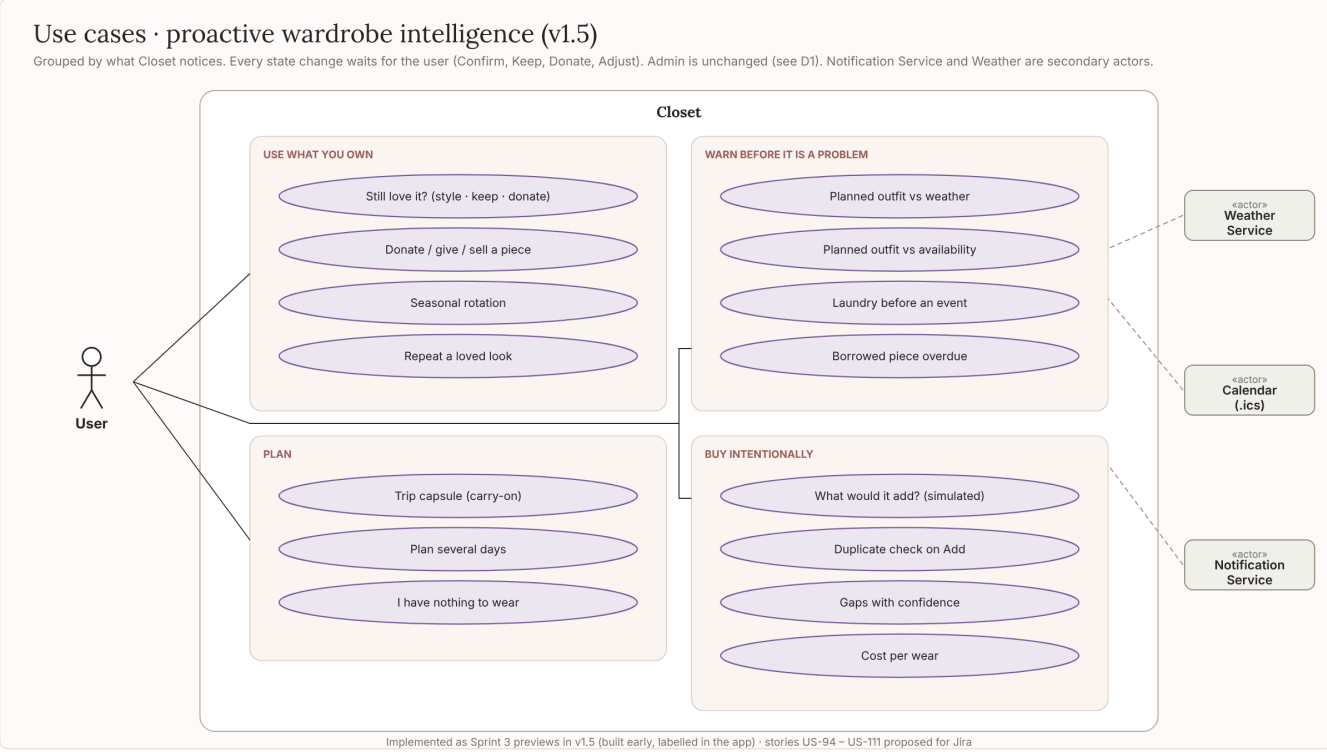


Figure R4-1. Use cases · proactive wardrobe intelligence (v1.5). The overview D1 — User, Admin and the external services — is unchanged and remains in Appendix B.

UC	Name	Primary actor	Secondary actors	Requirements
UC-23	Still love it? (style · keep · donate)	User	—	FR-81, FR-82
UC-24	Donate / give / sell a piece	User	—	FR-83

UC	Name	Primary actor	Secondary actors	Requirements
UC-25	Rotate the wardrobe for the season	User	Weather Service	FR-84
UC-26	Monitor planned outfits	User	Weather Service, Calendar	FR-85, FR-89
UC-27	Plan days / pack a trip capsule	User	Weather Service	FR-86, FR-95
UC-28	Buy intentionally (impact, duplicates, gaps)	User	—	FR-92 – FR-94
UC-29	Remember and repeat outfits	User	—	FR-87, FR-88
UC-30	Receive proactive notices	User	Notification Service	FR-98

Admin (Milestone 1 feedback): present in D1 with UC-19 – UC-20 (accounts, system health). The Admin has no access to wardrobes, prices or donations (FR-77).

R4.5 Traceability for the new requirements

Req.	Story	Screen	API	Module	Data	Tests
FR-81	US-94	Home “Still love it?” · Analytics → Still love it?	GET /intelligence/forgotten	Wardrobe Insights (insights.forgottenCandidates) · Proactive service	wardrobe.items · availability.item_status · intelligence.snoozes	IN-01 – IN-08 · API-PR01
FR-82	US-94	Still love it? card	POST /intelligence/forgotten/:itemId/keep · POST /recommendations {fixedId}	Proactive service · Recommendation Engine	intelligence.snoozes	IN-06 · API-PR01 · API-PR91 · E2E s06–s07
FR-83	US-95	Donate this item? · piece page	POST/DELETE /items/:id/rehome · POST /items/:id/rehome/done	Wardrobe · Smart Shopping (ItemDeleted observer)	wardrobe.rehoming · wardrobe.deletions	API-PR02 · API-PR03 · API-PR92 · OF-07 · E2E s08–s09
FR-84	US-96	Analytics → Seasonal rotation · Home notice	GET/POST /intelligence/seasonal	Wardrobe Insights · Availability state machine	availability.item_status · intelligence.snoozes	IN-10 – IN-12 · API-PR10
FR-85	US-97	Home “Closet noticed” · Notification Center	GET /intelligence/laundry-risk	Wardrobe Insights (laundryRisk)	availability.item_status · planner.events	IN-20 – IN-22 · API-PR20
FR-86	US-97	Ask Closet “plan my next 4 days”	POST /intelligence/plan-days	Wardrobe Insights (planDays) · Recommendation Engine	—	IN-23 · API-PR21
FR-87	US-98	Ask Closet “Can I repeat yesterday’s outfit?”	POST /intelligence/repeat	Wardrobe Insights (repeatInsight)	outfits.wears · profile.profiles	IN-30 – IN-32 · API-PR62
FR-88	US-99	Analytics → Outfit history · Ask Closet	POST /outfits/wear · GET /intelligence/memory	Outfits · Wardrobe Insights (memoryMatches, memorySignals) · Engine	outfits.wears	IN-33 – IN-34 · API-PR60
FR-89	US-100	Home “Closet noticed” · Planner	GET /intelligence/monitor · POST /intelligence/monitor/:eventId/adjust · POST /intelligence/monitor/ignore	Proactive monitor · Planner · Recommendation Engine (swap)	planner.events (planned_temp_c, planned_condition)	IN-40 – IN-42 · API-PR30 – PR32 · E2E s14
FR-90	US-101	Piece page · Home · Ask Closet	GET /intelligence/borrowed · POST /intelligence/borrowed/:itemId/remind · PATCH /items/:id/status	Proactive service · Availability	availability.item_status (note, until) · intelligence.snoozes	API-PR40
FR-91	US-102	Analytics → Wardrobe health · Ask Closet	GET /intelligence/health	Closet Intelligence (health) · Wardrobe Insights (explainHealth)	—	WH-01 – WH-05 · IN-80
FR-92	US-103	Add a piece → Review	POST /items/similar	Wardrobe · Smart Shopping (similarity)	—	DU-01 – DU-02 · API-PR70
FR-93	US-104	Closet gaps · Smart Shopping	GET /intelligence/gaps	Wardrobe Insights (gapConfidence)	—	IN-60 – IN-61 · API-PR72

Req.	Story	Screen	API	Module	Data	Tests
FR-94	US-105	Shopping → Do I need this? What would it add?	POST /shopping/impact	Smart Shopping · Wardrobe Insights (purchaseImpact) · Engine	—	IN-62 – IN-63 · API-PR71
FR-95	US-106	Planner → Packing	POST /packing/plan · POST /packing/trips · /pack · /unpack	Packing · Wardrobe Insights (tripCapsule) · Availability	packing.trips	IN-70 – IN-73 · API-PR80
FR-96	US-107	Ask Closet	POST /chat (event_rescue)	Ask Closet · Recommendation Engine · gaps	—	AI-40 · API-PR93
FR-97	US-108	Piece page · Analytics → Cost per wear	PATCH /items/:id · GET /intelligence/value	Wardrobe · Wardrobe Insights (costPerWear, valueInsights)	wardrobe.items (price, currency, purchase_date)	IN-50 – IN-52 · API-PR50
FR-98	US-109	Notification Center · Home “Closet noticed”	GET /notifications · POST /notifications/read	Closet Intelligence (notifications)	intelligence.notifications_read · intelligence.snoozes	API-PR31 · API-PR40 · E2E s01
FR-99	US-110	Ask Closet	POST /chat · POST /chat/undo/:id	Ask Closet (keyword parser / LLM adapter)	assistant.actions	AI-40 – AI-47 (63 cases) · API-PR90 – PR93 · E2E s10
FR-100	US-19 (changed)	Style me · Another chance	POST /recommendations	Recommendation Engine (pickDiverse)	—	DV-01 – DV-07 · demo:check
FR-101	US-26 (changed)	Style me · Try-on · Lookbook	POST /outfits · PATCH /outfits/:id	Outfits (outfitKey, mergeSave)	outfits.outfits	API-PR61 · OF-05 · E2E s05

Full matrix with Sprint 1 rows: `05_Testing/Closet_Traceability_Matrix_v1.5.xlsx`. Tests ran 497/497 passing in the final run (see the Test Plan and Report v1.2 for how they were executed).

R4.6 Appendix D addendum · change log C-70 – C-84

ID	Type	Section	Old → new
C-70	Added	4.21	Proactive wardrobe intelligence: FR-81 – FR-99 (built early as Sprint 3 previews in v1.5).
C-71	Changed	4.7 FR-78 / US-19	“no two share more than 2 pieces” → FR-100: at most one shared MAJOR garment; diversity-aware ranking.
C-72	Changed	4.12 FR-43 / US-26	Saved look identity (FR-101): Save + Favourite never create two records.
C-73	Changed	4.14 FR-51	“13 14 whitelisted functions” → 31; questions read state; out-of-scope answer (FR-99).
C-74	Changed	4.16 FR-60	Purchase analysis counts every owned piece (laundry included); distinct outfits only (FR-94).
C-75	Changed	4.3 FR-15 · FR-71	Delete reasons: + Donated, Given away, Sold / rehomed — only through FR-83, never a replacement (BR-12).
C-76	Added	5.4	NFR-16 determinism, NFR-17 one source of truth for status, NFR-18 offline fallback parity.
C-77	Added	5.5	BR-12, BR-13.
C-78	Changed	4.18 FR-64 – FR-66	Packing Mode → trip capsule with luggage limit, reuse and remove/re-plan (FR-95).

ID	Type	Section	Old → new
C-79	Changed	4.17 FR-62 – FR-63	Notification Center: types, priorities, dedupe keys, snoozes, ≤ 2 on Home (FR-98).
C-80	Added	3.3	GET /vocabulary now also returns STATUS_TRANSITIONS; the web app no longer has its own copy.
C-81	Added	6.3	Data: migration 006 (price, currency, purchase_date, wardrobe.rehoming, intelligence.snoozes, wears.event_title / note, events.planned_temp_c / planned_condition, profiles.allow_repeats).
C-82	Added	Appendix B	Use-case diagram D19 (proactive intelligence) added; D1 (with Admin) unchanged.
C-83	Changed	6.4	Traceability extended to FR-81 – FR-101 (Closet_Traceability_Matrix_v1.5.xlsx).
C-84	Changed	AS-3	Without an API key the keyword parser answers the same 31 functions.

Proposed stories (not yet in Jira)

ID	Title	Priority	Estimate	Requirements
US-94	Still love it? — style, keep or donate a long-unworn piece	Medium	3 pts	FR-81, FR-82
US-95	Donate, give or sell a piece	Medium	3 pts	FR-83
US-96	Seasonal rotation	Low	2 pts	FR-84
US-97	Laundry before an event and multi-day plans	Medium	3 pts	FR-85, FR-86
US-98	Repeat a look — intentionally	Low	2 pts	FR-87
US-99	Outfit memory	High	3 pts	FR-88
US-100	Planned-outfit monitor (weather + availability)	High	3 pts	FR-89
US-101	Borrow tracker	Low	2 pts	FR-90
US-102	Explain my Wardrobe Health	Low	1 pts	FR-91
US-103	Duplicate check when adding	High	2 pts	FR-92
US-104	Gaps with confidence	High	2 pts	FR-93
US-105	What would this purchase add?	High	3 pts	FR-94
US-106	Trip capsule	Medium	5 pts	FR-95
US-107	I have nothing to wear	Medium	2 pts	FR-96
US-108	Cost per wear	Low	2 pts	FR-97
US-109	One notification center for proactive notices	Medium	2 pts	FR-98
US-110	Ask Closet understands my wardrobe questions	High	5 pts	FR-99
US-111	Offline demo on real logic	Medium	3 pts	NFR-18

> The baseline v1.1 revision 3 specification follows.

Software Requirements Specification

for

Closet

Version 1.1 · revision 3 (final)

Prepared by Habiba Khashaba, Salma, Bashar, Karim and Youssef

Team Closet · CSCE 3701 Software Engineering · The American University in Cairo

10/26/2026

Document version	1.1 (revision 3)
Milestone	2 — Sprint 1 review
Project build referenced	Closet application v1.4

Repository: github.com/HabibaKhashaba/closet · Jira: aucegypt-team-r73i3oon.atlassian.net (project SCRUM)

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Appendix A: Glossary

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Revision History

Name	Date	Reason For Changes	Version
Habiba Khashaba (team)	Sep 2026	Milestone 0: problem, users, first feature list	0.1
Habiba Khashaba (team)	10/13/2026	Milestone 1 baseline: requirements and architecture in one combined specification	1.0
Habiba Khashaba (team)	10/25/2026	Milestone 2. The requirements are now a separate SRS that follows the IEEE / Wiegers template; the design moved to the SDD (Closet_SDD_v1.1). It includes every change from the interim Milestone 2 drafts: the Admin actor, authentication and authorization requirements, US-61 moved to Sprint 2, the customization asked for in the Milestone 1 review, and the v1.3 product requirements (explainable recommendations, Closet has learned, Wardrobe Health, conflict detection, occasion templates, outfit history, duplicate warning, notifications, packing, activity log). Each change is listed in Appendix D.	1.1
Habiba Khashaba (team)	10/26/2026	Milestone 2, revision 2 after the team review: the core product flow is stated first (three looks → change a piece → virtual try-on → feedback → learning → shopping); deletion reasons and their effect on shopping and learning are explicit requirements (FR-15, FR-58, FR-71 – FR-73, BR-10); the shopping list with its sources and “add to my shopping list” through Ask Closet (FR-61, FR-74); recommendation presentation and try-on flow (FR-75, FR-76); the Admin’s privacy boundary (FR-77). Changes C-50 – C-57.	1.1 (rev. 2)
Habiba Khashaba (team)	10/26/2026	Revision 3 (final pass for Milestone 2): virtual try-on on the user’s own photo with privacy (FR-42, FR-80, NFR-05); three looks with distinct names (FR-78); change one piece from up to three alternatives with their new match % (FR-34); style a forgotten piece — every look contains it (FR-35); Suggest a fix in Mix & Match (FR-79). Change log C-64 – C-67. This document describes application build v1.4.	1.1 rev. 3

How changes are tracked. Text added after the Milestone 1 baseline (version 1.0) is highlighted green; text removed is ~~struck through in red~~. Every change also has an ID (C-01 – C-40, C-50 – C-57) in Appendix D. A change of release or scope is always shown as a ~~old~~ new pair, so a reader can see what moved.

1. Introduction

1.1 Purpose

This document specifies the software requirements for **Closet, release 1.0** (the product delivered on December 1, 2026 for CSCE 3701 Software Engineering, Fall 2026, at the American University in Cairo). It covers the whole product: the web application, its server, and its use of external services. It states *what* the system must do and how well. *How* it is designed is described in the companion Software Design Description, Closet_SDD_v1.1.

Every requirement carries the sprint that delivers it. Sprint 1 (October 11 – 24) is complete and is graded at Milestone 2; requirements marked Sprint 2, Sprint 3 or Stretch are planned, even where a working preview already exists.

1.2 Document Conventions

- **Requirement IDs.** Functional requirements are FR-01 – FR-80, numbered across the document and grouped by system feature in section 4. Non-functional requirements are NFR-01 – NFR-15. User stories (US-xx) and use cases (UC-xx) keep the IDs they had in version 1.0, so references in Jira, the tests and the SDD stay valid.
- **Wording.** “Shall” is a binding requirement. Each requirement is written to be verifiable; its test is named in the traceability matrix (section 6.4).
- **Priority.** Every feature has a MoSCoW priority (Must / Should / Could) inherited by its detailed requirements unless a requirement states its own.
- **Release.** S1 = Sprint 1 (implemented), S2 = Sprint 2, S3 = Sprint 3, Stretch = only if time allows, Future = not in this course. “Built early” means a working preview exists before its sprint; it is accepted only at that sprint’s review and is not counted as Sprint 1 work.
- **Vocabulary.** Terms in *italics* are defined in Appendix A. Allowed values (dress codes, statuses, categories) are fixed in Appendix A and used unchanged by the client, server and database.

1.3 Intended Audience and Reading Suggestions

Reader	Read first	Then
Instructor and TA (product owners)	1.4 Product Scope, 2.2 Product Functions	4 System Features, 6.1 Release plan
Developers (Salma, Bashar, Karim)	2 Overall Description, 4 System Features	3 External Interfaces, the SDD
Tester (Youssef)	4 (each “Functional Requirements” list), 5	6.4 Traceability, Appendix B
Scrum lead and documentation (Habiba)	Revision History, Appendix D	6.1 – 6.3 Backlog

Section 2 describes the product and its users; section 3 the external interfaces; section 4 the 20 system features with their functional requirements; section 5 the quality, security and safety requirements; section 6 the release plan, the product backlog with acceptance criteria, and traceability. The appendices hold the glossary, the analysis models (use case diagrams, use case descriptions, state model), the TBD list and the change log.

1.4 Product Scope

The problem. People lose time every morning deciding what to wear, wear a small part of their wardrobe again and again, forget clothes they own, and buy things they do not need, which costs money and adds to textile waste.

The product. Closet is an AI-assisted digital wardrobe. It knows which clothes a user owns and which are in the laundry, lent out or packed. For an occasion it returns three complete outfits that fit the dress code, the user's style and modesty preference, and the weather, and it explains why each one works. The user can change one piece, try the look on a stylised avatar, wear it, save it and rate it. An AI chat, Ask Closet, answers questions about the closet and acts on it. Closet learns from every reaction, shows what it has learned in plain words, and lets the user confirm or undo it.

Why it is more than an outfit generator. Six kinds of context — *my wardrobe, my style, my weather, my calendar, my history, my feedback* — feed one system whose results are better outfits, smarter shopping and better use of what the user owns. The product loop is:

> **Wardrobe → Context → Recommendation → Feedback → Learning → Better recommendation**

Goals. (1) A decision in under a minute every morning. (2) Higher wardrobe utilisation. (3) Fewer duplicate or unnecessary purchases. (4) Transparent, controllable AI.

Out of scope for release 1.0: photo-real try-on on the user's own photo, Arabic interface, bulk upload, social sharing, any purchasing or payment, and live two-way calendar sync.

1.5 References

Reference	Version / date	Location
Closet Software Design Description (SDD)	1.1, 10/25/2026	Closet_SDD_v1.1.pdf (this package)
Closet Milestone 2 Sprint 1 report and test report	v1.3, 10/25/2026	Closet_M2_Test_Report_v1.3.pdf
Product backlog and sprint board	live	Jira: aucegypt-team-r73i3oon.atlassian.net/jira/software/projects/SCRUM/boards/1
Source code, tests, migrations	commit history	github.com/HabibaKhashaba/closet
K. Wiegers, SRS template	1999	course material
IEEE Std 830-1998 / ISO/IEC/IEEE 29148	—	requirements practice
R. Pressman, *Software Engineering: A Practitioner's Approach*	course slides	Requirements Engineering, Architectures
Egypt Personal Data Protection Law 151/2020	2020	privacy principles (NFR-05)
WCAG 2.1	AA	accessibility target (NFR-07)

2. Overall Description

2.1 Product Perspective

Closet is a new, self-contained product. It is a web application (phone and laptop browsers) backed by one server and one database, and it uses six external services through adapters: a hosted **language model** (interpret chat messages), a **weather API** (Open-Meteo), **calendar** files from Google, Outlook or Apple, a **vision** service (tag suggestions, stretch), a **virtual try-on** service (future; the app draws a stylised avatar today) and a **notification** service (future; notices are in-app today). The product must keep working when any external service is unavailable.

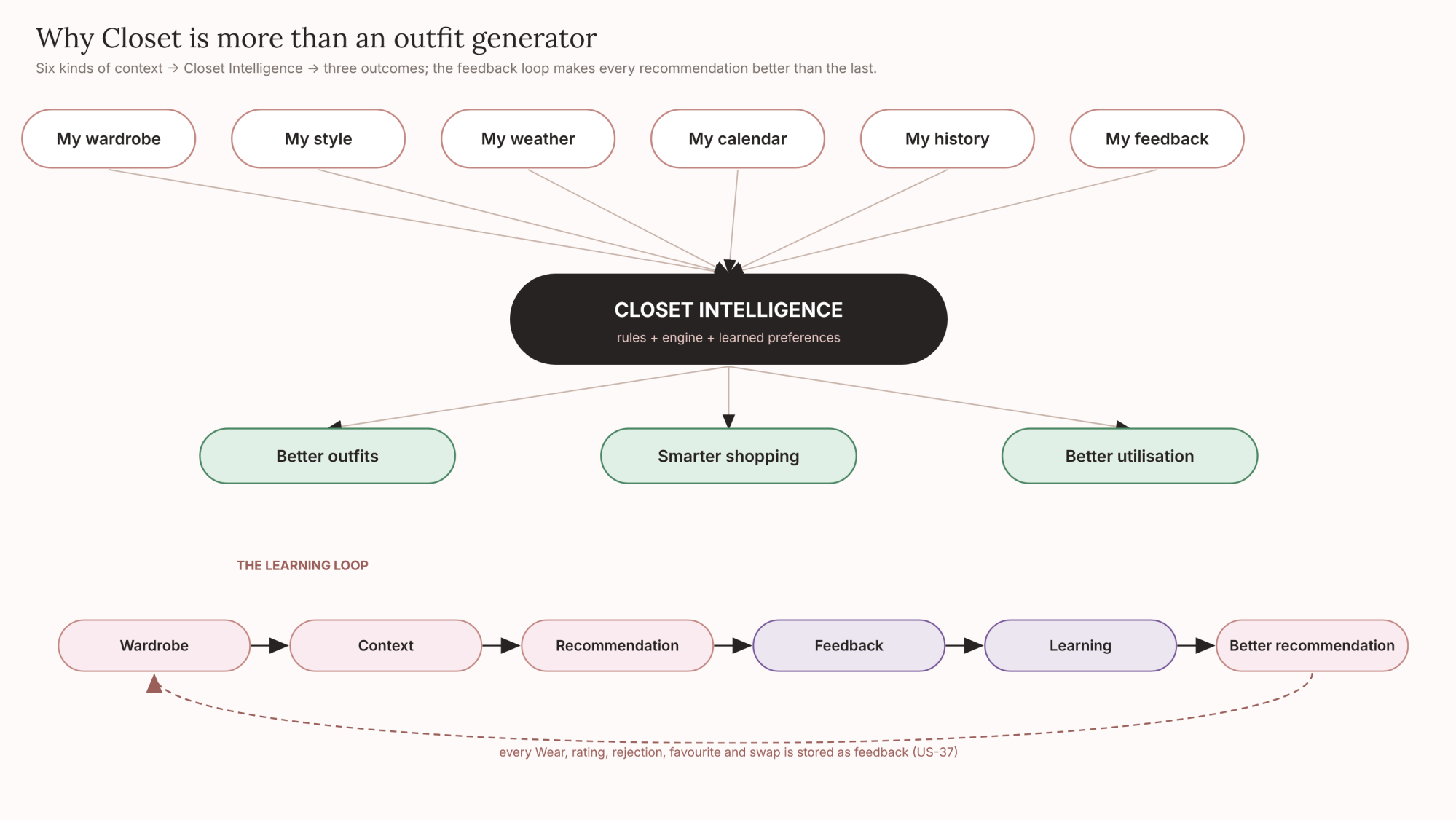


Figure 2-1 – Product perspective: the context Closet uses and what it produces.

The heart of Closet. Closet is not a wardrobe list with extras. Its core is one loop the user repeats every day: **choose the occasion → Closet generates three complete looks → change any piece → try the look on in the dressing room → wear, save, favourite, rate or reject it with a reason**. Every one of those actions is feedback; Closet learns from it, and from the reasons pieces leave the closet, and uses what it learned in the next three looks and in the shopping list (Figure 2-2). The closet, availability, laundry and planner exist so that this loop works on real, clean, available clothes.

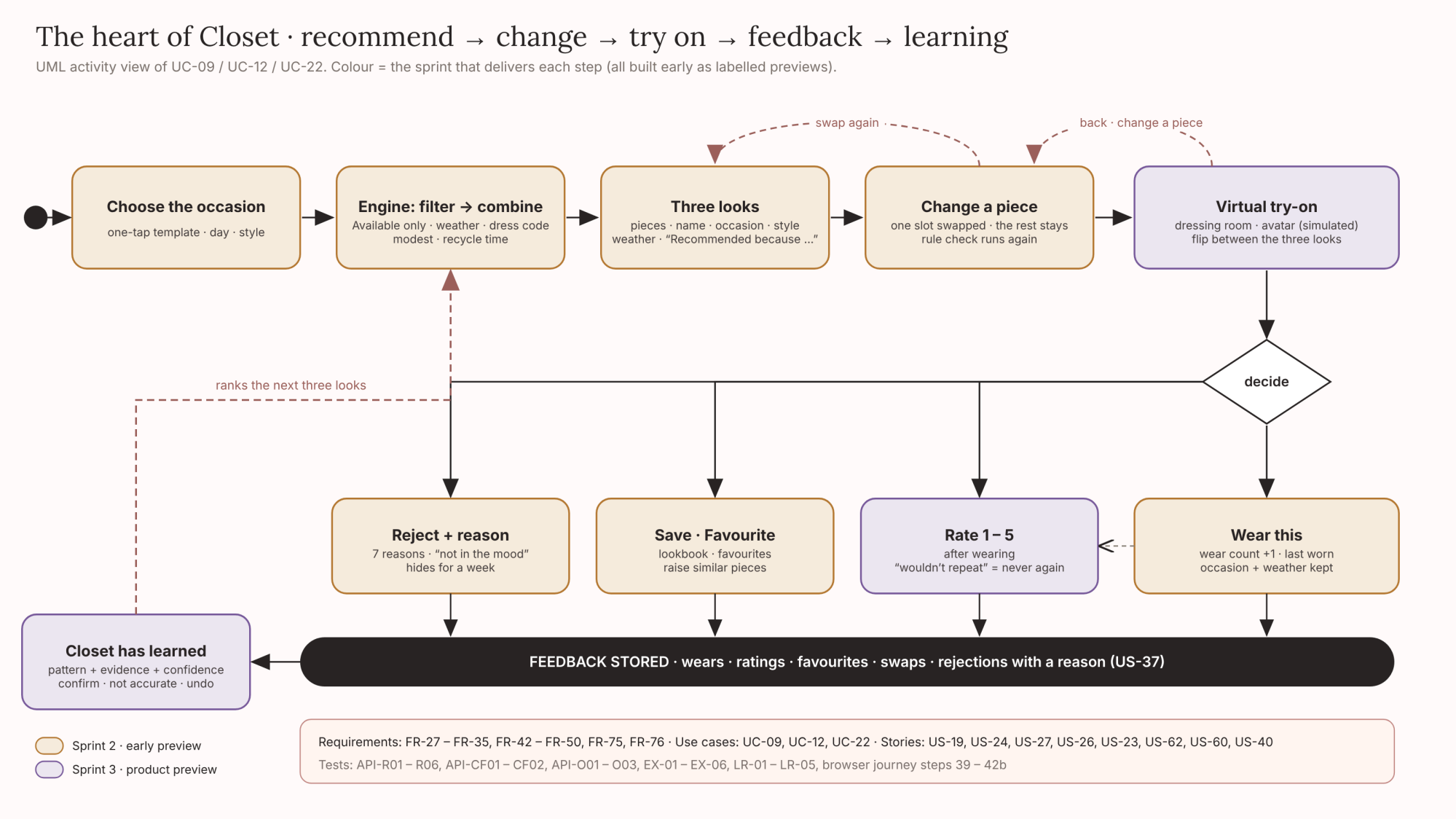


Figure 2-2 – The core product flow (UML activity view of UC-09, UC-12 and UC-22). Sprint 2 and Sprint 3 steps are built early as labelled previews.

2.2 Product Functions

- **Account and security:** register, sign in (email or Google), sign out, account lockout, delete account and all data. Admin role with account management.
- **Style profile and wardrobe rules:** style quiz, dress code, modest preference, city; laundry basket time, laundry day, recycle time.
- **Digital closet:** add pieces by photo with details, browse and filter, edit, delete with a reason (the reason drives replacement shopping and learning), closet gaps.
- **Availability and laundry:** five statuses, laundry basket, Wash done, return dates.
- **Planner:** events in a calendar, calendar import, outfits planned for events, conflict warnings.
- **Outfits (the core):** three explained recommendations, change one piece, virtual try-on from every look, save, favourite, wear, rate, reject with a reason; style a forgotten piece, Mix & Match builder, occasion templates, outfit history.
- **Ask Closet:** AI chat that answers and acts through whitelisted functions, with Undo.
- **Insights:** wear analytics, forgotten pieces, Wardrobe Health, “Closet has learned”.
- **Smart shopping:** what the closet really needs, “Do I need this?”, duplicate-purchase warning, a shopping list fed by Ask Closet, replacements, gaps and suggestions — never by pieces let go as “no longer my style”.
- **Notifications and activity:** one Notification Center; the user’s own activity log.
- **Packing (stretch):** plan and pack a suitcase for a trip.

2.3 User Classes and Characteristics

User class	Characteristics	Importance
University students (primary user)	Daily use on the phone; laundry-aware; budget-conscious; dress for class, presentations, dinners	Most important
Young professionals	Correct dress codes for work and interviews; plan for events	Important
People with large wardrobes	Need to remember and rotate what they own	Important
Users who need modest options	Covered shoulders and longer hemlines respected everywhere	Important (must never be violated)
Admin	A team member who runs the service: manages accounts, sees service health and totals. Never sees anyone’s photos, pieces, outfits or chats.	Operational

The User and the Admin are the two primary actors in the use case model (Appendix B, Figure B-1).

2.4 Operating Environment

- Client: current Chrome (Android, desktop) and Safari (iOS, macOS), from 360 px wide; responsive up to a 1440 px presentation laptop.
- Server: Node.js 22 on Linux (Render web service or equivalent); PostgreSQL 16.
- Network: HTTPS only. External services are reached over HTTPS through adapters.

2.5 Design and Implementation Constraints

- Course constraints: 5 team members, 280 hours of capacity (80 h per sprint), three two-week sprints and a release week; deadline December 1, 2026.
- Technology chosen by the team: React (web), Node.js + Express (server), PostgreSQL (data), hosted AI APIs only (no model training).
- Privacy: designed with Egypt’s PDPL (Law 151/2020) principles: consent, minimal data, export, deletion.

- AI use: hosted language models may only choose among whitelisted functions; they never receive photos or direct database access.
- Secrets (database URL, token keys, API keys) live only in environment variables, never in the repository.
- Free or student tiers only for external services (NFR-12).

2.6 User Documentation

In-app help (Ask Closet answers “help”), short empty-state explanations on every screen, the README in the repository (setup, demo account, demo reset), and the presenter pack and demo script for the Milestone 2 review.

2.7 Assumptions and Dependencies

ID	Assumption or dependency	If wrong
AS-1	Users photograph pieces themselves on a plain background	Tiles look noisy; the vision stretch feature becomes more valuable
AS-2	Open-Meteo stays free and reachable	The weather adapter falls back to a seasonal estimate and says so
AS-3	A language-model API key is available for the demo	Ask Closet uses the keyword parser; same ~~13~~ 14 functions
AS-4	Users export calendars as .ics files	Manual events still work (Sprint 1)
AS-5	Rule-based scoring is good enough for release 1.0	The Strategy design allows a learned scorer later (SDD 3.3)
AS-6	One region and one time zone per user (Africa/Cairo by default)	Calendar times need a per-user time zone (TBD-3)

3. External Interface Requirements

3.1 User Interfaces

- **UI-1.** A single-page web app with a left navigation on wide screens and a bottom tab bar (Home, Outfits, Ask, Closet, More) on phones. Every screen carries a label of the sprint that delivers it: “Sprint 1 · Implemented”, “Sprint 2 · Early preview”, “Sprint 3 · Product preview” or “Stretch · Concept preview”.
- **UI-2.** Every screen has one primary goal and a physical metaphor that matches it: the closet as rails, the laundry as a basket, Mix & Match as a styling table, the try-on as a dressing room, saved looks as a lookbook, packing as a suitcase.
- **UI-3.** Every list or screen has a written empty state, a loading state, and success messages; destructive actions ask for confirmation; reversible actions offer Undo.
- **UI-4.** Form errors appear next to the field in plain words; the server never shows raw errors, stack traces or JSON to the user.
- **UI-5.** Scores are always labelled with what they are (“match %”, “rule score”, “confidence”) and can be expanded to show how they were computed.
- Screen images are in the SDD, section 6.2.

3.2 Hardware Interfaces

The phone or laptop camera and file picker are used through the browser for photos (JPG, PNG, WebP, HEIC up to 10 MB). No other hardware is required. A camera inside the closet was considered in Milestone 0 and replaced by availability statuses.

3.3 Software Interfaces

ID	External software	Data in / out	Purpose	Release
SI-1	PostgreSQL 16	All structured data	Storage	S1
SI-2	Open-Meteo forecast and geocoding API	City, date → temperature, condition	Weather-aware outfits, planner, packing	S2 (built early)
SI-3	Hosted language model (Anthropic or OpenAI)	Message text → one whitelisted function + arguments	Ask Closet	S3 (built early)
SI-4	Calendar files (.ics, RFC 5545) from Google, Outlook, Apple	Title, date, time	Import events	S2 (built early)
SI-5	Vision API	Photo → suggested tags with confidence	AI pre-fill	Stretch
SI-6	Virtual try-on service	Outfit → photo-real image	Try-on on the user’s photo	Future
SI-7	Push / e-mail notification service	Notice text	Reminders outside the app	Future
SI-8	Google Identity Services	ID token → verified email	Sign in with Google	S1

3.4 Communications Interfaces

- **CI-1.** All client–server traffic is HTTPS (TLS 1.2+), JSON over REST under /api/v1, documented as one API contract (SDD 3.2).
- **CI-2.** Authenticated requests carry a bearer token; tokens expire after 2 hours and end on sign-out or suspension.
- **CI-3.** Photos are served only through signed URLs that expire.
- **CI-4.** Calls to external services time out (weather 4 s, language model 8 s) and fall back (NFR-06).

4. System Features

The functional requirements are organised by system feature. Each feature lists its priority, its stimulus/response sequence, and its requirements; the user stories and acceptance criteria behind each requirement are in section 6.2.

4.1 Authentication and account

4.1.1 Description and Priority. Users create an account with email and password or Google, sign in and out, and can delete the account with all its data. Accounts are protected against guessing and against access by other users. Priority: **Must** · release: **S1**.

4.1.2 Stimulus/Response Sequences.

- User enters email and password → the system creates the account and opens the style quiz.
- User signs in with a wrong password → the system shows one generic error and counts the failure; after 5 failures the account is locked for 15 minutes.
- User signs out → the session ends on every device.
- User asks to delete the account, types the confirmation phrase and the password → all data and photos are removed.

4.1.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-01	The system shall let a user register with a unique email and a password of 8 – 72 characters, or with a verified Google account.	US-01	S1
FR-02	The system shall sign a user in only with correct credentials and shall show the same message for an unknown email and a wrong password.	US-01	S1
FR-03	The system shall lock an account for 15 minutes after 5 consecutive failed sign-ins.	US-01	S1
FR-04	The system shall end a session after 2 hours, on sign-out, and when the account is suspended.	US-01, US-80	S1
FR-05	The system shall return only the signed-in user’s data; a request for another user’s record shall behave as if the record does not exist.	US-01	S1
FR-06	The system shall delete the account, every piece, photo, outfit, event and log entry after a double confirmation (phrase and password).	US-06	S1

4.2 Style profile and wardrobe rules

4.2.1 Description and Priority. A short style quiz and an editable profile tell Closet the user’s styles, colours, occasions, usual dress code, modest preference and city. Wardrobe rules (asked for in the Milestone 1 review) customise laundry and repetition. Priority: **Must** · release: **S1 / S2**.

4.2.2 Stimulus/Response Sequences.

- First sign-in → the quiz opens (≤ 4 steps, skippable).
- User changes a setting → the next recommendation uses it.

4.2.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-07	The system shall collect styles, colours, occasions, dress code (1 – 5), modest preference and city in at most four steps and shall allow skipping.	US-02 – US-04	S1
FR-08	The system shall let the user edit every profile value at any time.	US-05	S2

ID	Requirement	Stories	Release
FR-09	The system shall let the user set a laundry basket time (1 – 14 days), automatic return, a laundry day, and a recycle time (0 – 30 days); out-of-range values shall be refused with a field message.	US-84, US-85	S2 (built early)

4.3 Digital closet

4.3.1 Description and Priority. Every piece is a photo plus structured details from one shared vocabulary. The closet can be browsed like a wardrobe (rails per category) and filtered. Priority: **Must** · release: **S1**.

4.3.2 Stimulus/Response Sequences.

- User takes or uploads a photo, fills the details and reviews them → the piece is saved as Available.
- User filters by category and status or searches → the list and counts update.

4.3.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-10	The system shall accept a JPG, PNG, WebP or HEIC photo up to 10 MB, check its type from the file content, remove location data, and store it privately.	US-08	S1
FR-11	The system shall require a category, a colour and at least one dress-code level, and shall accept only values from the vocabulary (Appendix A).	US-09	S1
FR-12	The system shall show one review screen and create the piece only when the user saves it.	US-11	S1
FR-13	The system shall list pieces grouped by category and allow combining category, status and text filters with live counts.	US-12	S1
FR-14	The system shall let the user edit any detail of a piece later.	US-71	S1
FR-15	The system shall ~~delete a piece only with one of six reasons and shall keep the reason (not the photo) for smart shopping.~~ require the user to provide a deletion reason when deleting a clothing item — Torn / damaged, No longer my style, Became too small or Other — and shall keep the reason, category, colour and styles, never the photo.	US-13	S2 (built early)
FR-71	The system shall use deletion reasons to update shopping suggestions and personalisation behaviour exactly as defined in BR-10, and shall tell the user the effect when the piece is deleted.	US-13, US-33, US-34	S2 / S3 (built early)

4.4 Availability and laundry

4.4.1 Description and Priority. Each piece has exactly one of five statuses. The laundry basket holds the pieces in the wash; “Wash done” returns them all. Priority: **Must** · release: **S1**.

4.4.2 Stimulus/Response Sequences.

- User moves a piece to In laundry → it appears in the basket and leaves every recommendation.
- User presses Wash done → every piece in the basket becomes Available and the basket shows “All clean”.

4.4.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-16	The system shall give every piece exactly one status (Available, Laundry, Borrowed, Packed, Unavailable) and allow only the 10 moves of Figure B-3, refusing any other move with an explanation.	US-56	S1
FR-17	The system shall list every piece in the laundry with the date it went in and the date it is expected back.	US-15, US-84	S1
FR-18	The system shall return every Laundry piece to Available with one action and report how many returned.	US-16	S1

ID	Requirement	Stories	Release
FR-19	With automatic return on, the system shall make a Laundry piece Available once its basket time has passed.	US-84	S2 (built early)
FR-20	The system shall store an optional return date (not in the past) and a note of up to 60 characters for Borrowed, Packed and Unavailable pieces.	US-86	S2 (built early)

4.5 Closet gaps

4.5.1 Description and Priority. Shows how many pieces exist per outfit group and dress-code level, and where the closet is thin. Priority: **Could** · release: **S1**.

4.5.2 Stimulus/Response Sequences.

- User opens Closet gaps → a table of groups × levels with the gaps highlighted.

4.5.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-21	The system shall report a gap wherever a group (tops and dresses, bottoms and layers, shoes) has one piece or fewer at a level the user dresses for.	US-64	S1

4.6 Planner and calendar

4.6.1 Description and Priority. Events in a month calendar, imported from calendar files, with outfits planned for them and the wardrobe's own dates (laundry day, pieces due back) on the same calendar. Priority: **Must** · release: **S1 / S2 / S3**.

4.6.2 Stimulus/Response Sequences.

- User adds an event with title, date, time and occasion → it appears in the month view.
- User imports an .ics file → future events of the next year are added once, with a guessed occasion.
- User plans an outfit for an event → the outfit appears on the event and on Home.

4.6.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-22	The system shall store events with a title (1 – 80 characters), date, time and one occasion, and show them in a month view.	US-46	S1
FR-23	The system shall import events from an .ics file (Google, Outlook, Apple), skipping past events, events more than a year ahead and duplicates.	US-87	S2 (built early)
FR-24	The system shall mark laundry day, pieces back from the wash, borrowed pieces due back and packed pieces returning on the calendar.	US-87	S2 (built early)
FR-25	The system shall propose three outfits for an event using its dress code, that day's forecast and the pieces that will be available on that day.	US-47	S3
FR-26	The system shall let the user choose one outfit as the planned outfit for an event.	US-48	S3

4.7 Outfit recommendations and explanations

4.7.1 Description and Priority. For an occasion (and optionally a style and a day) Closet returns three complete, different outfits from Available pieces that fit the dress code, the modest preference and the weather, each explained. Priority: **Must** · release: **S2**.

4.7.2 Stimulus/Response Sequences.

- User chooses an occasion → three looks, each with a match %, its four part scores and "Why Closet chose it".

- User changes one piece → only that slot changes.
- User opens Try on from a look → the dressing room shows it on the avatar; the user flips between the three looks, then wears, saves or favourites it, or goes back to change a piece.
- User asks to style a forgotten piece → three looks that all contain it.

4.7.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-27	The system shall return three complete outfits in ≤ 2 s, no two sharing more than two pieces when the closet allows.	US-19	S2
FR-28	The system shall never include a piece that is not Available on the outfit's date (BR-1).	US-17	S2
FR-29	The system shall use the forecast for the user's city and the chosen day: no warm piece at ≥ 26 °C, a warm layer below 19 °C, and a note at ≥ 38 °C or ≤ 8 °C.	US-20	S2
FR-30	The system shall use only pieces that suit the occasion's dress code and shall respect the modest preference everywhere.	US-21	S2
FR-31	The system shall not suggest a top, bottom or dress worn within the user's recycle time.	US-85	S2 (built early)
FR-32	The system shall show, for every recommended outfit, a "Recommended because …" sentence and the factors that shaped it, naming only facts the system actually used: occasion and dress code, style preference, modesty, weather, availability, recent wear history, favourites and prior rejection patterns.	US-57	S2 (built early)
FR-33	The system shall show a match % that is the average of four defined part scores (colour, style, formality, weather; Appendix A) and the rule score, and shall be able to show how both are computed.	US-57, US-43	S2
FR-34	The system shall offer up to three Available alternatives for one slot of an outfit, each with the recalculated match %, and shall replace exactly one piece of an outfit with the ~~best Available alternative for that slot~~alternative the user picks and store the swap as feedback.	US-24	S2
FR-35	The system shall build three outfits around one chosen piece — including a forgotten piece — every outfit containing that piece; the piece must be Available, and the weather and recycle-time filters shall not remove it.	US-59	S2
FR-78	The system shall give each of the three outfits its own name, never two alike in one set.	US-19	S2 (built early)
FR-75	The system shall present each of the three outfits visually — its pieces, name, occasion, dress code, style, weather suitability and explanation — and shall offer on each: change a piece, virtual try-on, save, favourite, wear, rate after wearing, and reject with a reason; after a piece is changed the updated outfit is shown with its updated scores.	US-19, US-24, US-26, US-27, US-23, US-60, US-62	S2 (built early)

4.8 Occasion templates

4.8.1 Description and Priority. One tap fills the occasion, dress code and style for common situations so that asking for an outfit is faster. Priority: **Should** · release: **S2**.

4.8.2 Stimulus/Response Sequences.

- User taps "Interview" → the outfits use Interview, Business Casual and a classic style; the user can still change each value.

4.8.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-36	The system shall offer nine occasion templates — University, Work, Dinner, Wedding, Casual outing, Formal event, Travel, Gym, Interview — each setting an occasion, a dress-code level, a style and a weather sensitivity, all editable afterwards.	US-90	S2 (built early)

4.9 Conflict detection

4.9.1 Description and Priority. Domain rules that warn when a planned outfit can no longer be worn. Priority: **Could** · release: **S2**.

4.9.2 Stimulus/Response Sequences.

- User moves a planned piece to the laundry → Home, the Planner and the Notification Center show “Your navy blazer is in the laundry … Choose a replacement?” with the best replacement.
- User accepts → only that piece is replaced and the warning disappears.

4.9.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-37	The system shall detect, for events in the next 30 days, every planned piece that is in Laundry, Borrowed, Packed or Unavailable and will not be back by the event, and shall propose the best Available replacement for that slot.	US-79	S2 (built early)
FR-38	The system shall flag a piece planned for two events less than 4 hours apart on the same day.	US-79	S2 (built early)
FR-39	The system shall replace only the conflicting piece when the user accepts the replacement, and refuse if the conflict no longer exists.	US-79	S2 (built early)

4.10 Mix & Match (outfit builder)

4.10.1 Description and Priority. The user lays up to four pieces on a styling table and Closet checks the look live. Priority: **Must** · release: **S2**.

4.10.2 Stimulus/Response Sequences.

- User picks pieces slot by slot → a match %, part scores, the five rules with pass / fail, one explaining paragraph.
- User asks for a fix → Closet swaps the weakest piece, up to three times.

4.10.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-40	The system shall offer only Available pieces per slot by default (top or dress, bottom or outerwear, shoes, bag or accessory).	US-42	S2
FR-41	The system shall show the rule score with pass / fail for the five criteria and one sentence naming the failing criterion.	US-43, US-66	S2
FR-79	The system shall offer “Suggest a fix” in Mix & Match: apply the single swap that most improves the rule score (then the match %), at most three times, and show the match % before and after.	US-66	S2 (built early)

4.11 Virtual try-on

4.11.1 Description and Priority. ~~A stylised avatar wears the look in a dressing room~~The user sees the look on a full-body photo of herself (a clearly labelled preview; the stylised avatar is the fallback) so the user can compare looks before dressing. Priority: **Should** · release: **S3**.

4.11.2 Stimulus/Response Sequences.

- User presses Try on → chooses a full-body photo → every piece is placed on the photo; the user flips between the three looks, adjusts the fit and can remove the photo; without a photo → the avatar wears every piece in its colours~~; the user flips between looks and changes skin tone~~.

4.11.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-42	The system shall let the user choose a full-body photo of herself (explicit selection; JPG, PNG or WebP ≤ 15 MB) and show the chosen outfit placed on that photo, labelled “Virtual Try-On Preview · simulated on your photo”; when no photo is chosen it shall draw every piece of the outfit on a stylised avatar in ≤ 1 s~~, without any photo of the user,~~ and ~~shall~~ label it as a stylised preview. When a photo-real try-on provider is unavailable, the preview shall remain usable (graceful fallback).	US-27 – US-29	S3 (built early)
FR-76	The system shall open the virtual try-on from any recommended outfit (including one with a changed piece), let the user move between the three looks, and offer wear, save, favourite and “change a piece”, returning to the same three looks.	US-27	S3 (built early)
FR-80	The system shall keep the try-on photo only on the user’s device for the browser session, shall never send it to the server, and shall delete it when the user chooses “Remove my photo” or closes the tab.	US-27	S3 (built early)

4.12 Saved outfits and outfit history

4.12.1 Description and Priority. Saved looks form a lookbook; every “Wear this” builds an outfit history. Priority: **Must** · release: **S2**.

4.12.2 Stimulus/Response Sequences.

- User saves or favourites a look → it appears in the lookbook.
- User presses Wear this → wear counts update and the outfit appears in the history with its occasion and weather.

4.12.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-43	The system shall save outfits and mark favourites, and shall warn that a saved outfit with a non-Available piece cannot be worn today.	US-26, US-51	S2
FR-44	The system shall add 1 to the wear count and set last-worn to today for every piece of a worn outfit, and record the occasion, the temperature and the condition of that day.	US-23, US-91	S2
FR-45	The system shall list worn outfits newest first with date, occasion, pieces, rating, weather and each piece’s wear count, filterable by date range, occasion and minimum rating.	US-91	S2 (built early)
FR-46	The system shall let the user rate a worn outfit from 1 to 5.	US-62	S3

4.13 Feedback and learning (“Closet has learned”)

4.13.1 Description and Priority. Every reaction is stored as feedback. Closet detects patterns, shows them with evidence and confidence, and changes the profile only when the user confirms. Priority: **Must** · release: **S2 / S3**.

4.13.2 Stimulus/Response Sequences.

- User rejects a look and gives a reason → stored; “Not in the mood today” hides that look for 7 days.
- Closet shows “You often reject looks as too formal · 3 looks · 60 % confidence” → Confirm lowers the usual dress code; Not accurate dismisses it; Undo restores the previous value exactly.

4.13.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-47	The system shall store wears, swaps, rejections with one of seven reasons, favourites and hides, each with its time and pieces.	US-37, US-60, US-61	S2
FR-48	The system shall detect patterns from rejections, favourites and wear history and show each with the detected pattern, its evidence count and a confidence, and the profile change that confirming would make.	US-40, US-63	S3 (built early)
FR-49	The system shall change the profile only when the user confirms a pattern, shall never show a dismissed pattern again, and shall restore the previous values exactly on Undo.	US-40	S3 (built early)
FR-50	The system shall use confirmed preferences and repeated rejection reasons when it ranks future outfits.	US-60, US-63	S3 (built early)
FR-73	When two or more items deleted as “No longer my style” share a style, the system shall show that as a learned pattern with its evidence; once the user confirms it, pieces of that style shall rank lower and the style is removed from the profile, and Undo restores both exactly.	US-34, US-40	S3 (built early)

4.14 Ask Closet (AI chat agent)

4.14.1 Description and Priority. A chat that knows the user’s closet, laundry, calendar and weather, answers in plain words, and acts through Closet functions. Priority: **Must** · release: **S3**.

4.14.2 Stimulus/Response Sequences.

- User asks “Do I need another white top?” → Closet answers from the closet (“Probably not. You already own 2 similar white tops”) and shows the similar pieces.
- User says “Put my jeans in the laundry” → the status changes and Undo is offered.

4.14.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-51	The system shall map each message to exactly one of ~~13~~ 14 whitelisted functions, run it only on the user’s own data, and never name a piece the user does not own.	US-31, US-76	S3 (built early)
FR-52	The system shall list the function it used and offer Undo for every change it made.	US-76	S3 (built early)
FR-53	The system shall answer with the keyword parser when no language model is configured, it fails, or it takes more than 8 s.	US-31	S3 (built early)
FR-74	The system shall let the user add an item to the shopping list through Ask Closet — “put beige trousers on my shopping list”, “add this to my shopping list” after a “Do I need …?” question, “add a replacement for my torn loafers” — tagged “Added by Ask Closet” (and “Replacement” when it replaces a piece), with the similar-item warning and Undo.	US-36	S3 (built early)

4.15 Wear analytics, forgotten pieces and Wardrobe Health

4.15.1 Description and Priority. An editorial view of what the user really wears, what is forgotten, and how healthy the wardrobe is — each with an action. Priority: **Must** · release: **S3**.

4.15.2 Stimulus/Response Sequences.

- User opens Analytics → utilisation, pieces worn this month, forgotten pieces, heavy rotation, most and least worn, the Wardrobe Health score with its parts.

4.15.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-54	The system shall show utilisation (pieces worn in the last 30 days ÷ pieces owned) and the most and least worn pieces.	US-52, US-69	S3

ID	Requirement	Stories	Release
FR-55	The system shall list forgotten pieces (worn before, not in the last 40 days) with the days since last worn and a “Give it another chance” action that builds three outfits around the piece.	US-54	S3
FR-56	The system shall compute a Wardrobe Health score from 0 to 100 using the published formula (Appendix A), show every part and penalty with its points, and give up to three actions; an empty closet shall score 0 with an explanation.	US-89	S3 (built early)

4.16 Smart shopping and duplicate warning

4.16.1 Description and Priority. A curated board of what the closet really needs, each item answering “Why do I need this?”, and a check before buying. Priority: **Must** · release: **S3**.

4.16.2 Stimulus/Response Sequences.

- User opens Smart shopping → the shopping bag (Ask Closet items, replacements, gaps) and gaps and missing basics, each with its reason and how many new outfits it unlocks.
- User deletes worn-out loafers as “Torn / damaged” → a replacement appears in the shopping bag; a hoodie deleted as “No longer my style” never comes back as a suggestion.
- User checks “white top” → “Skip · You already own 2 similar white tops” with the pieces and their similarity.

4.16.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-57	The system shall suggest ~~replacements for well-worn pieces removed as too small, too big or damaged, pieces for real gaps, and missing basics,~~ pieces for real gaps and missing basics, each with a reason, the number of new complete outfits it makes and the owned pieces it pairs with.	US-32, US-33	S3 (built early)
FR-72	The system shall recommend a replacement item for every item deleted as “Torn / damaged” (same category and colour) and a next-size replacement for every item deleted as “Became too small”, by adding it to the shopping list tagged “Replacement”.	US-33	S3 (built early)
FR-58	The system shall ~~never suggest a category and colour the user removed as “no longer my style”.~~ avoid treating items deleted as “No longer my style” as shopping replacements: it shall never add or suggest a piece of the same category and colour, and shall answer “Think twice” when the user checks one.	US-34	S3 (built early)
FR-59	The system shall answer “Do I need this?” with one verdict (Good buy, Optional, Skip, Think twice, Not yet) and its reason.	US-88	S3 (built early)
FR-60	The system shall warn before a purchase when the user owns pieces similar to the one considered, state how many (“You already own 2 similar white tops”), show each with its similarity and what matched (category, colour, style, formality, season), and offer Buy anyway, Ask Closet and View similar items.	US-65	S3 (built early)
FR-61	The system shall keep a shopping list ~~(add, remove, bought) that persists.~~ whose items show their source — Added by Ask Closet, Replacement, Wardrobe gap, Smart suggestion or Added by you — and a warning when a similar piece is already owned; the user can add, mark bought and remove items, and the list persists (BR-11).	US-36	S3 (built early)

4.17 Notification Center

4.17.1 Description and Priority. One place for every notice from the planner, availability, packing, analytics and weather. Priority: **Could** · release: **S3**.

4.17.2 Stimulus/Response Sequences.

- User opens the bell → notices newest and most important first, unread marked, at most three highlighted today.

4.17.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-62	The system shall notify the user of: an event tomorrow without a planned outfit, a planned piece that became unavailable, a borrowed piece due back, laundry day tomorrow, rain expected for an event, a packing list not yet packed for a trip starting within three days, Wash done, and a forgotten piece.	US-92	S3 (built early)
FR-63	The system shall keep read / unread state per notice and highlight at most three unread notices a day (NFR-13).	US-92, US-74	S3 (built early)

4.18 Packing mode

4.18.1 Description and Priority. Plan a suitcase for a trip from the destination forecast, the style profile, available pieces and planned events. Priority: **Could** · release: **Stretch**.

4.18.2 Stimulus/Response Sequences.

- User enters Paris, 4 days, Travel, activities → a suitcase with pieces, outfits per day, counts, missing outfits and weather warnings.
- User presses Mark as packed → every listed piece becomes Packed until the day after the trip.

4.18.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-64	The system shall propose one outfit per trip day from Available pieces, re-using pieces across days, using each day's destination forecast and any planned event that day, and report pieces, outfits, days and missing outfits.	US-68	Stretch (preview)
FR-65	The system shall warn about rain without a suitable outer layer, cold without a warm layer, heat, and missing live forecasts.	US-68	Stretch (preview)
FR-66	The system shall set every listed Available piece to Packed with a return date the day after the trip, report pieces it could not pack, and exclude packed pieces from normal recommendations.	US-68	Stretch (preview)

4.19 Administration

4.19.1 Description and Priority. A small Admin experience, separated from the user role, that manages accounts and shows the service's health without exposing wardrobe contents. Priority: **Should** · release: **S3**.

4.19.2 Stimulus/Response Sequences.

- Admin opens Admin → accounts with status and piece counts; Suspend ends that user's sessions.
- Admin opens System health → database, weather, language model, try-on and notification services with their status.

4.19.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-67	The system shall let an Admin list accounts (email, role, status, created, piece count) and suspend or restore an account; an Admin shall not be able to suspend their own account.	US-80	S3 (built early)
FR-68	The system shall refuse every administration function to a non-admin (403).	US-80	S3 (built early)
FR-77	The system shall give an Admin no route to any user's pieces, photos, outfits, shopping list or chats; the Admin sees account records and totals only (BR-3).	US-80	S3 (built early)
FR-69	The system shall show service health (database, weather, language model, try-on, notifications), uptime and the count of unexpected errors, and read-only configuration lists — never any user's photos, pieces, outfits or chats.	US-81, US-83	Stretch (preview)

4.20 Activity log

4.20.1 Description and Priority. Users can see what happened to their account and closet — useful for security, for support and for explaining state changes. Priority: **Should** · release: **S2**.

4.20.2 Stimulus/Response Sequences.

- User opens Profile → recent activity: “Signed in”, “Moved to Laundry · Navy blazer”, “Wash done · 4 pieces”, “Piece deleted · Black straight jeans · Became too small”.

4.20.3 Functional Requirements.

ID	Requirement	Stories	Release
FR-70	The system shall record each successful change of NFR-15 with its time and the piece’s name, and show the user their last 50 entries newest first; reads shall not be recorded and no user shall see another user’s log.	US-93	S2 (built early)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement	Verification
NFR-01	Three outfits in ≤ 2 s (95th percentile, closet ≤ 300 pieces); chat reply ≤ 8 s (90th percentile); avatar drawn in ≤ 1 s on the phone; Wardrobe Health and Notification Center in ≤ 1 s for 300 pieces	Load test with a 300-piece closet
NFR-02	Tag suggestions optional, ≤ 10 s; nothing saved without user confirmation	Test with the vision service slow and down

5.2 Safety Requirements

ID	Requirement	Verification
NFR-14	No action deletes or changes data without the user's intent: account deletion needs the phrase and the password; deleting a piece needs a reason; a learned preference changes the profile only after Confirm and can be undone exactly; Ask Closet changes can be undone	API tests API-D*, API-LR01, API-AC04
—	Closet comments only on clothes, never on bodies or sizes; extreme heat (≥ 38 °C) and cold (≤ 8 °C) produce a note	Review; engine tests

5.3 Security Requirements

ID	Requirement	Verification
NFR-04	HTTPS only; ~~hashed passwords~~ bcrypt-hashed passwords (8 – 72 characters); login tokens ~~checked at the gateway~~ (signed, 2 h) checked on every protected route and ended by sign-out; every query scoped to the user (another user's data returns 404); lockout after 5 failed logins for 15 minutes; admin functions need the admin role (403 otherwise); security headers, CORS limited to the web app, rate limits on sign-in, validated input, generic error messages; secrets only in environment variables	Automated security tests (auth, ownership, admin)
NFR-05	Privacy: consent, minimal data, export, deletion within 24 h; only message text and garment photos (no location data, no faces) leave our servers; photo location data removed on upload; the Admin sees counts and account records only, never photos, pieces, outfits or chats; the virtual try-on photo never leaves the user's device (kept for the browser session only, removable at any time)	Privacy checklist each sprint; deletion and admin tests
NFR-15	Auditability: every successful change a user makes (sign-in, sign-out, profile change, piece added / edited / deleted, status change, Wash done, event added, outfit planned, packing, learned-preference decision, admin action) is recorded in that user's activity log with its time; reads are not logged; a user sees only their own log	API-N02, API-N03

5.4 Software Quality Attributes

ID	Attribute	Requirement	Verification
NFR-03	Usability	A new user finishes the quiz and adds a first piece in < 3 min; core tasks ≤ 3 taps; every screen has an empty state, a loading state and success messages; Undo for reversible actions	Test with 5 users; browser journey
NFR-06	Reliability	If the language model, weather or vision service fails, the app keeps working: keyword parser, seasonal weather estimate, manual tags; never a crash	Tests with each adapter forced to fail

ID	Attribute	Requirement	Verification
NFR-07	Accessibility	WCAG 2.1 AA target: contrast ≥ 4.5 : 1, touch targets ≥ 44 px, visible keyboard focus, every control reachable by keyboard, meaningful alt text on garment images, labels for screen readers, never colour alone (sprint labels and statuses also carry text), reduced motion respected	Automated audit + manual check
NFR-08	Portability	Current Chrome and Safari, from 360 px to 1440 px wide	Test on 4 devices or emulators
NFR-09	Testability	≥ 60 % line coverage of core modules; every Must acceptance criterion has a test; tests run on every pull request	Coverage report (test report)
NFR-10	Explainability	Every outfit shows why it was chosen, naming only facts the system used; every score is labelled; every learned lesson is listed in plain words with its evidence and confidence	UI review; EX tests
NFR-11	Localisation readiness	All text in one place; layout supports right-to-left; °C and EGP	Code review
NFR-12	Cost	AI calls cached and capped per user per day; free or student tiers	Weekly usage check
NFR-13	Notifications	At most 3 highlighted notices per user per day	Unit / API test

5.5 Business Rules

ID	Rule
BR-1	Only *Available* pieces are ever used in a recommended, planned or packed outfit. A piece that will be back by the date of a future event counts as available for that date.
BR-2	A piece has exactly one status; only the 10 allowed status moves are possible (Appendix B, Figure B-3).
BR-3	A user can see and change only their own data. An Admin can manage accounts but never see wardrobe contents. An Admin cannot suspend their own account.
BR-4	With the modest option on, no non-modest piece appears anywhere.
BR-5	“Not in the mood today” hides a look for 7 days without counting as a dislike.
BR-6	A learned pattern changes the profile only when the user confirms it; the previous value is kept so Undo restores it exactly.
BR-7	Smart shopping never suggests re-buying a category and colour the user removed as “no longer my style”.
BR-8	A suspended account cannot sign in and all its sessions end.
BR-9	Every score shown is computed by a defined rule (Appendix A); nothing is presented as a trained-model prediction.
BR-10	Deletion reasons and their effects: **Torn / damaged** → a replacement (same category and colour) is added to the shopping list; **Became too small** → a replacement in the next size is added; **No longer my style** → never a replacement, never suggested again (same category and colour), and when two or more such pieces share a style Closet offers to learn it (BR-6); **Other** → the reason is recorded only. The photo is never kept.
BR-11	A shopping-list item shows where it came from (Added by Ask Closet, Replacement, Wardrobe gap, Smart suggestion, Added by you) and warns when the user already owns a similar piece (similarity ≥ 70).

6. Other Requirements

6.1 Release plan and scope

We commit to fully building ~~50~~ ~~55~~ 62 of the ~~70~~ 84 stories, including all 43 Must stories, by December 1, 2026. Sprint 1 scope did not change after sprint planning except US-61, which moved to Sprint 2 (C-04). Stories added after Milestone 1 are Sprint 2 or Sprint 3 stories built early as previews, or stretch goals.

Sprint	Dates	Goal	Committed stories
S1 ✓	Oct 11 – 24	A working closet	US-01, 02, 03, 04, 06, 08, 09, 11, 12, 71, 15, 16, 56, ~~61,~~ 64, 46 (15 — all done, 169 tests pass)
S2	Oct 25 – Nov 7	Outfits that explain themselves	US-05, 13, 17, 19, 20, 21, 57, 23, 24, 26, 37, 42, 43, 66, 51, 59, 60, 61, 84, 85, 86, 87, 79, 90, 91, 93
S3	Nov 8 – 21	AI chat, learning, planning, analytics, admin	US-27, 31, 76, 32, 33, 34, 36, 65, 62, 38, 39, 40, 47, 48, 52, 69, 54, 80, 88, 89, 92
Release	Nov 22 – Dec 1	Integration, testing, demo	Real language-model adapter on; end-to-end tests of every Must criterion

Stretch (16): US-10 · US-72 · US-58 · US-28 · US-29 · US-63 · US-75 · US-44 · US-45 · US-50 · US-67 · US-53 · US-70 · US-74 · US-68 packing (working concept preview) · US-81 admin system health (working preview). ~~US-79~~ and ~~US-65~~ left the stretch list (C-28, C-29).

Future (6): US-07 Arabic interface · US-14 bulk upload · US-18 laundry reminder · US-30 photo-real try-on · US-82 admin privacy requests · US-83 admin configuration (read-only preview exists).

6.2 Product backlog: user stories and acceptance criteria

Each story reads “As a user, I want to …” (“As an admin …” for US-80 – US-83). Release: S1 – S3 = committed sprint, Stretch = only if time allows, Future = not in this course. Every acceptance criterion is a test (section 6.4).

Backlog summary. 84 user stories in 10 epics, prioritised with MoSCoW; every Must story is committed for December 1.

Epic	Stories	Must	Should	Could	Committed	Stretch	Future
E1 Accounts & Profile	~~7~~ ~~11~~ 12	6	~~0~~ ~~1~~ 2	~~1~~ 4	~~6~~ ~~7~~ 8	~~0~~ 1	~~1~~ ~~4~~ 3
E2 Digital Closet	9	6	1	2	6	2	1
E3 Availability & Laundry	~~5~~ 8	3	~~1~~ 4	1	~~4~~ 7	0	1
E4 Outfit Recommender	~~9~~ 10	7	~~2~~ 3	0	~~8~~ 9	1	0
E5 Virtual Try-On	4	0	1	3	1	2	1
E6 Ask Closet AI Agent	~~8~~ 9	5	1	~~2~~ 3	~~8~~ 9	~~1~~ 0	0
E7 Feedback & Learning	9	6	2	1	7	2	0
E8 Outfit Builder	5	3	1	1	3	2	0
E9 Plan Ahead	~~7~~ 9	3	~~1~~ 2	~~3~~ 4	~~3~~ 6	3	~~1~~ 0
E10 Saved & Wear Analytics	~~7~~ 9	4	~~1~~ 2	~~2~~ 3	~~4~~ 6	3	0
Total	**~~70~~ 84**	**43**	**~~11~~ 19**	**~~16~~ 22**	**~~50~~ 62**	**~~15~~ 16**	**~~5~~ 6**

E1 Accounts & Profile

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-01	Register and log in with email or Google	Must	S1	UC-01	Password ≥ 8 chars or Google; wrong password shows an error; no other user's data is ever returned; logout ends the session; 5 failed sign-ins lock the account for 15 minutes
US-02	Answer a short style quiz: styles, colours, occasions	Must	S1	UC-02	≤ 4 steps, skippable; answers saved to the profile (S1); answers change the first recommendation (verified in S2 with US-19)
US-03	Set my usual dress code and a modest preference	Must	S1	UC-02	One of the 5 levels; every piece stores a modest flag (S1); with modest on, no non-modest piece appears anywhere (verified in S2 with US-21)
US-04	Set my city so outfits use my local weather	Must	S1	UC-02	City saved to the profile (S1); forecast on Home is for that city (verified in S2 with US-20)
US-05	Edit my profile and settings at any time	Must	S2	UC-02	Every quiz answer is editable; the next recommendation uses it
US-06	Delete my account and all my photos and data	Must	S1	UC-01	Double confirmation; all data and photos deleted within 24 h
US-07	Use the app in Arabic (RTL)	Could	Future	—	Out of scope
US-80	(Admin) Suspend or reactivate a user account	Should	S3	UC-15	Search by email; sees status and counts, never photos; suspended user cannot sign in; every admin action logged; non-admin gets 403
US-81	(Admin) See system health and usage totals	Could	~~Future~~ Stretch (preview)	UC-16	Uptime, error rate, fallback counts; totals only
US-82	(Admin) Complete a user's export or deletion request	Could	Future	UC-17	Same deletion as US-06; recorded in the audit log
US-83	(Admin) Manage app lists and feature switches	Could	Future	UC-18	Categories, dress codes, occasions; switch the AI chat off when the language model is down
US-93	See my account activity	Should	S2 (built early)	UC-01	Sign-ins, profile changes, pieces added / edited / deleted, status changes, Wash done, events, planned outfits, packing and learned-preference decisions listed newest first with the time; reads are not logged; only the owner sees it

E2 Digital Closet

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-08	Add a piece by taking or uploading a photo	Must	S1	UC-03	JPG/PNG/WebP/HEIC ≤ 10 MB (type checked from the file content; location data removed) stored and shown as the piece's image
US-09	Enter category, colour, fabric, season, size, style, dress code	Must	S1	UC-03	Values from Appendix A; category, colour and ≥ 1 dress-code level required
US-10	Let AI suggest those details from the photo	Should	Stretch	UC-03	Suggestions ≤ 10 s with confidence; nothing saved until confirmed; failure leaves a usable empty form
US-11	Review and edit every detail before saving	Must	S1	UC-03	One review screen; the piece exists only after Save

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-12	Browse and filter my closet by category and status	Must	S1	UC-04	Filters combine; counts update
US-71	Edit a piece’s details later	Must	S1	UC-04	Changes appear in all outfits and filters
US-72	Enter an optional purchase price	Could	Stretch	UC-04	Optional, ≥ 0, in EGP
US-13	Delete a piece with a reason	Must	S2	UC-05	Impossible without one of the 4 reasons; piece removed from saved and planned outfits
US-14	Upload several pieces at once	Could	Future	—	Out of scope

E3 Availability & Laundry

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-56	Set a status per piece (5 statuses)	Must	S1	UC-06	Exactly one status, shown as a badge
US-15	Put a piece in the laundry basket and take it out	Must	S1	UC-06	One tap each way; basket lists all Laundry pieces
US-16	Return everything with one “Wash done” tap	Should	S1	UC-06	All Laundry pieces become Available
US-84	Set my laundry basket time and laundry day	Should	S2 (built early)	UC-06, UC-19	1 – 14 days; each Laundry piece shows when it is back; with auto-return on, it becomes Available after that time (logged); laundry day shown on the calendar with a reminder the day before
US-86	Add a return date and note to a borrowed, packed or unavailable piece	Should	S2 (built early)	UC-06	Date not in the past; note ≤ 60 characters (“Nour”); shown on the piece, on the calendar and in reminders; cleared when the piece is Available
US-85	Set a recycle time so the same clothes are not suggested again too soon	Should	S2 (built early)	UC-19, UC-09	0 – 30 days; a top, bottom or dress worn in the last N days is not suggested; shoes, bags and accessories may repeat; 0 = no limit
US-17	Never get an outfit with an unavailable piece	Must	S2	UC-09	Test: 100 generated outfits with random statuses contain 0 non-Available pieces
US-18	Reminder for pieces long in the laundry	Could	Future	—	Out of scope

E4 Outfit Recommender

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-19	Choose an occasion and a style and get 3 outfits of 4 pieces	Must	S2	UC-09	3 outfits in ≤ 2 s; each complete; no two share more than 2 pieces when the closet allows
US-20	Have the weather shape the outfits	Must	S2	UC-09	≥ 26 °C: no warm piece; < 19 °C: a warm layer; ≥ 38 °C or ≤ 8 °C: heat or cold note
US-21	Filter by dress code and modest option	Must	S2	UC-09	Every main piece suits the level; modest respected

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-57	See why each outfit works	Must	S2	UC-09	3–5 reason lines and a “rule score” label on every outfit; a “Recommended because …” sentence and factors that name only facts the engine used (occasion and dress code, style, modesty, weather, availability, wear history, favourites, rejections); a match % with its four parts
US-23	Tap “Wear this” to log the outfit and its pieces	Must	S2	UC-12	Wear count +1 and last-worn = today for all 4 pieces; analytics update
US-24	Change one of the 4 pieces, the other 3 stay	Must	S2	UC-09	Only that slot changes; alternatives are Available; swap stored as feedback
US-26	Save an outfit, and mark it as a favorite	Must	S2	UC-09	Saved list; favorite = a starred saved outfit
US-59	“Style this piece”: 3 outfits with one piece fixed	Should	S2	UC-08	All 3 outfits contain that piece
US-58	Choose repeats: don’t mind / avoid last 7 days / something different	Should	Stretch	UC-09	“Avoid 7 days” uses no piece worn in the last 7 days when an alternative exists
US-90	Use one-tap occasion templates	Should	S2 (built early)	UC-09	9 templates (University, Work, Dinner, Wedding, Casual outing, Formal event, Travel, Gym, Interview) set occasion, dress code and style; every value can still be changed

E5 Virtual Try-On

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-27	See an outfit on a stylised avatar	Should	S3	UC-09	All 4 pieces drawn in their colours in ≤ 1 s on the phone; no photo of the user
US-28	Flip between the 3 looks in the try-on	Could	Stretch	UC-09	Swipe switches looks
US-29	Choose the avatar’s skin tone	Could	Stretch	UC-09	≥ 6 tones, saved in profile
US-30	Try on with a photo of myself	Could	Future	—	Out of scope

E6 Ask Closet AI Agent

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-31	Chat with an agent that knows my closet, calendar and weather	Must	S3	UC-07	Reply ≤ 8 s (p90); never names a piece the user does not own
US-76	Have the agent act for me through Closet functions	Must	S3	UC-07	Only whitelisted functions; reply lists the function used; every change undoable
US-32	Ask “what does my closet need?”	Must	S3	UC-14	Item types listed, each with a reason
US-33	See torn or too-small pieces suggested for replacement	Must	S3	UC-14	Every Torn or Became-small deletion appears on the list

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-34	Never be told to re-buy “no longer my style” pieces	Must	S3	UC-14	No suggestion matches their category and colour
US-36	Keep a shopping list	Should	S3	UC-14	Add from chat or list, remove, persists
US-88	Ask “Do I need this?” before buying a piece	Could	S3 (built early)	UC-14	Verdict (Good buy, Optional, Skip, Think twice, Not yet) from similar pieces owned, new complete outfits it would make, and deletion reasons
US-64	See closet gaps by category and dress code	Could	S1	UC-14	Gap = ≤ 1 piece at a level the user dresses for
US-65	Get a duplicate-purchase warning	Could	~~S3 Stretch~~ S3 (built early)	UC-14	Same category and colour as an owned piece shows that piece; similarity 0 – 100 from category, colour, style, formality and season, ≥ 70 = similar; “You already own N similar …”; Buy anyway, Ask Closet, View similar items

E7 Feedback & Learning

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-37	Have wears, swaps and rejections stored as feedback	Must	S2	UC-09	Each event stored with time, outfit and pieces
US-60	Give a reason when I reject a look	Must	S2	UC-09	7 reasons; each changes exactly one preference
US-61	Hide a piece for 7 days without it being a dislike	Should	~~S1~~ S2	UC-09	Returns after 7 days; rejection count unchanged
US-62	Rate an outfit after wearing it	Must	S3	UC-12	Wouldn’t repeat = combination never recommended again
US-38	Be asked when I reject the same piece twice	Must	S3	UC-09	One in-app question at the 2nd rejection, within the daily limit
US-39	Answer: don’t like it / not these days / keep suggesting	Must	S3	UC-09	Exclude, pause 2 weeks, or reset
US-40	See what Closet learned in plain words, and undo it	Must	S3	~~UC-02~~ UC-22	One sentence per lesson; Undo restores the previous preference exactly; each pattern shows its evidence and confidence and asks Confirm / Not accurate; nothing changes before Confirm
US-63	Have favorites raise similar colours and styles	Could	Stretch	UC-09	Favoriting raises their weights
US-75	Pause learning, reset it, or download my data	Should	Stretch	UC-02	Pause stops updates; download gives a JSON file

E8 Outfit Builder

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-42	Pick a top/dress, bottom/layer, shoes and an extra	Must	S2	UC-11	Only Available pieces per slot

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-43	See a rule score with pass/fail for the 5 criteria	Must	S2	UC-11	Criteria and formula as in Appendix A
US-66	Read one plain sentence that explains a mismatch	Must	S2	UC-11	Names the failing criterion
US-44	Get one suggested swap that improves the score	Could	Stretch	UC-11	Applying it raises the score
US-45	Try on, save or wear a built outfit	Should	Stretch	UC-11	Same actions as a recommended outfit

E9 Plan Ahead

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-46	Add events by hand and see them in a calendar	Must	S1	UC-10	Title, date, time and one occasion; month view
US-87	Import my calendar and see wardrobe dates on it	Should	S2 (built early)	UC-10	.ics file from Google, Outlook or Apple; only future events in the next year; occasion guessed from the title; no duplicates; calendar marks laundry day, pieces back from the wash, borrowed pieces due, packed pieces; a future-date plan counts pieces back by then
US-47	Get 3 outfits for an event	Must	S3	UC-10	Uses the event's dress code, that day's forecast and Available pieces
US-48	Choose one as the planned outfit	Must	S3	UC-10	Shown on the event and on Home that day
US-50	Sync events from Google Calendar	Should	Stretch	UC-10	Read-only; next 30 days
US-67	Get a reminder the night before	Could	Stretch	UC-10	One in-app notice at 21:00, within the daily limit
US-79	Be warned if a planned piece becomes unavailable	Could	~~Stretch~~ S2 (built early)	UC-10	Warning with a one-tap swap; a piece back before the event is not a conflict; a piece planned for two events < 4 h apart is flagged
US-68	Packing assistant for trips	Could	~~Future~~ Stretch (preview)	— UC-20	~~Out of scope~~ Destination, dates, trip type, activities → one outfit per day re-using pieces; counts, missing outfits, weather warnings; Mark as packed sets each piece to Packed until the day after the trip
US-92	See every notice in one Notification Center	Could	S3 (built early)	UC-21	Conflicts, event tomorrow without an outfit, borrowed piece due, laundry day, rain for an event, packing not done, Wash done, forgotten piece; read / unread; at most 3 highlighted a day

E10 Saved & Wear Analytics

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-51	See saved outfits and filter favorites	Must	S2	UC-09	A saved outfit with a non-Available piece shows a warning and cannot be worn
US-52	See most and least worn pieces	Must	S3	UC-13	Top and bottom 5 by wear count
US-69	See my wardrobe utilization	Must	S3	UC-13	Formula in Appendix A; matches a manual count in tests

ID	I want to…	Priority	Release	UC	Acceptance criteria
US-54	See forgotten pieces	Must	S3	UC-13	Exactly the pieces matching Appendix A; each has “Style it”
US-53	See cost per wear for priced pieces	Could	Stretch	UC-13	Only priced pieces, with a note
US-70	Get a weekly forgotten-item challenge	Could	Stretch	UC-13	One per week, within the daily limit
US-74	Control notification types and the daily limit	Should	Stretch	UC-02	Types can be switched off; default 3 per day
US-89	See a Wardrobe Health score	Could	S3 (built early)	UC-13	0 – 100 from five parts minus two penalties (Appendix A), every part listed with its points; empty closet → 0 with a message; up to 3 actions
US-91	See my outfit history	Should	S2 (built early)	UC-13	Each “Wear this” shows date, occasion, pieces, rating, weather and each piece’s wear count; filter by date range, occasion and minimum rating

6.3 Data requirements

Each piece stores its owner, name, photo, category, colour, fabric, season, size, styles, dress-code levels, modest flag, status (with return date and note), wear count and last-worn date. Feedback events (wear with rating, occasion and weather; swap; rejection with reason; favourite; hide) are the only input to learning. Deletion reasons are kept after the piece is deleted, without the photo. Learned decisions keep the previous profile values for Undo. User data is never used to train models or shared with other users. All of a user’s data is removed when the account is deleted.

6.4 Requirements traceability

Every functional requirement traces to its use case, its user stories and the automated tests that verify it. The design side of the chain (component, API, data) is in the SDD, section 7, and the full matrix is a separate file in the package.

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-01	register with a unique email and a password of 8 – 72 characters, or…	UC-01	US-01	AT-01, API-A01 – A05, API-A16 – A18	S1
FR-02	sign a user in only with correct credentials and shall show the same…	UC-01	US-01	API-A06, API-A07, UT-S06	S1
FR-03	lock an account for 15 minutes after 5 consecutive failed sign-ins.	UC-01	US-01	API-A08 – A10	S1
FR-04	end a session after 2 hours, on sign-out, and when the account is…	UC-01	US-01, US-80	API-A11 – A15, UT-S01 – S05, API-AD03	S1
FR-05	return only the signed-in user’s data; a request for another user’s…	UC-01	US-01	API-W16 – W18, API-DL03, API-R06, API-PK03	S1
FR-06	delete the account, every piece, photo, outfit, event and log entry…	UC-01	US-06	AT-15, API-D01 – D03	S1

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-07	collect styles, colours, occasions, dress code (1 – 5), modest…	UC-02	US-02 – US-04	AT-02 – AT-04, API-P01 – P03	S1
FR-08	edit every profile value at any time.	UC-02	US-05	API-P04, API-P05	S2
FR-09	set a laundry basket time (1 – 14 days), automatic return, a laundry…	UC-19	US-84, US-85	API-C01 – C03	S2 (built early)
FR-10	accept a JPG, PNG, WebP or HEIC photo up to 10 MB, check its type…	UC-03	US-08	AT-05, API-W01 – W08, UT-P01 – P08	S1
FR-11	require a category, a colour and at least one dress-code level, and…	UC-03	US-09	AT-06, API-W09, API-W10, UT-V11 – V21	S1
FR-12	one review screen and create the piece only when the user saves it.	UC-03	US-11	AT-07	S1
FR-13	list pieces grouped by category and allow combining category, status…	UC-04	US-12	AT-08, API-W11 – W13	S1
FR-14	edit any detail of a piece later.	UC-04	US-71	AT-09, API-W14, API-W15	S1
FR-15	require the user to provide a deletion reason when deleting a…	UC-05	US-13	API-DL01 – DL03, API-DR05, DR-01	S2 (built early)
FR-71	use deletion reasons to update shopping suggestions and…	UC-05, UC-14	US-13, US-33, US-34	DR-01, DR-02, API-DR01 – DR05	S2 / S3 (built early)
FR-16	give every piece exactly one status (Available, Laundry, Borrowed,…	UC-06	US-56	AT-10, API-S01 – S06, ST-01 – ST-04	S1
FR-17	list every piece in the laundry with the date it went in and the date…	UC-06	US-15, US-84	AT-11, API-L01	S1
FR-18	return every Laundry piece to Available with one action and report…	UC-06	US-16	AT-12, API-L02 – L04	S1
FR-19	With automatic return on, the system shall make a Laundry piece…	UC-06, UC-19	US-84	API-C02, API-C03	S2 (built early)
FR-20	store an optional return date (not in the past) and a note of up to…	UC-06	US-86	API-C05	S2 (built early)
FR-21	report a gap wherever a group (tops and dresses, bottoms and layers,…	UC-14	US-64	AT-14, API-G01, UT-G01 – G07	S1
FR-22	store events with a title (1 – 80 characters), date, time and one…	UC-10	US-46	AT-13, API-E01 – E05, UT-V22 – V26	S1
FR-23	import events from an .ics file (Google, Outlook, Apple), skipping…	UC-10	US-87	API-PL03, CAL-01	S2 (built early)

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-24	mark laundry day, pieces back from the wash, borrowed pieces due back…	UC-10	US-87	API-PL01	S2 (built early)
FR-25	propose three outfits for an event using its dress code, that day's…	UC-10	US-47	API-C06, API-AC09	S3
FR-26	choose one outfit as the planned outfit for an event.	UC-10	US-48	API-K01, API-AC09	S3
FR-27	return three complete outfits in ≤ 2 s, no two sharing more than two…	UC-09	US-19	RE-09, API-R01	S2
FR-28	never include a piece that is not Available on the outfit's date (BR-1).	UC-09	US-17	RE-01, EX-04, API-PK02	S2
FR-29	use the forecast for the user's city and the chosen day: no warm…	UC-09	US-20	RE-05 – RE-08, WA-01 – WA-05	S2
FR-30	use only pieces that suit the occasion's dress code and shall respect…	UC-09	US-21	RE-02, RE-03, RE-24	S2
FR-31	not suggest a top, bottom or dress worn within the user's recycle time.	UC-09, UC-19	US-85	RE-21 – RE-23, API-C04	S2 (built early)
FR-32	show, for every recommended outfit, a "Recommended because …"…	UC-09	US-57	RE-12, EX-01 – EX-03, EX-06, API-EX01, API-EX02	S2 (built early)
FR-33	a match % that is the average of four defined part scores (colour,…	UC-09, UC-11	US-57, US-43	RE-10, RE-25 – RE-27, API-R05	S2
FR-34	offer up to three Available alternatives for one slot of an outfit,…	UC-09	US-24	RE-17, RE-18, RE-33, API-R04, API-CF04	S2
FR-35	build three outfits around one chosen piece — including a forgotten…	UC-08	US-59	RE-13, RE-14, RE-31, RE-32, API-R06, API-FP01, API-FP02	S2
FR-78	give each of the three outfits its own name, never two alike in one set.	UC-09	US-19	RE-30, API-CF03, API-R01, browser journey (40)	S2 (built early)
FR-75	present each of the three outfits visually — its pieces, name,…	UC-09, UC-12	US-19, US-24, US-26, US-27, US-23, US-60, US-62	API-CF01, API-CF02, API-R01, browser journey (39 – 41)	S2 (built early)
FR-36	offer nine occasion templates — University, Work, Dinner, Wedding,…	UC-09	US-90	OT-01, API-OT01	S2 (built early)
FR-37	detect, for events in the next 30 days, every planned piece that is…	UC-10	US-79	API-K01, API-K03	S2 (built early)
FR-38	flag a piece planned for two events less than 4 hours apart on the…	UC-10	US-79	API-K04	S2 (built early)

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-39	replace only the conflicting piece when the user accepts the…	UC-10	US-79	API-K02	S2 (built early)
FR-40	offer only Available pieces per slot by default (top or dress, bottom…	UC-11	US-42	RE-24, browser journey	S2
FR-41	the rule score with pass / fail for the five criteria and one…	UC-11	US-43, US-66	RE-19, RE-20, RE-29, API-R05	S2
FR-79	offer “Suggest a fix” in Mix & Match: apply the single swap that most…	UC-12	US-66	API-MM01, browser journey (44a – 44c)	S2 (built early)
FR-42	choose a full-body photo of herself (explicit selection; JPG, PNG or…	UC-09	US-27 – US-29	TO-01 – TO-03, browser journey (42a, 42, 42b)	S3 (built early)
FR-76	open the virtual try-on from any recommended outfit (including one…	UC-09	US-27	browser journey (42_tryon, 42b_tryon_next)	S3 (built early)
FR-80	keep the try-on photo only on the user’s device for the browser…	UC-09	US-27	browser journey (42: no request carries the photo)	S3 (built early)
FR-43	save outfits and mark favourites, and shall warn that a saved outfit…	UC-09	US-26, US-51	API-O01	S2
FR-44	add 1 to the wear count and set last-worn to today for every piece of…	UC-12	US-23, US-91	API-O02, API-OH01	S2
FR-45	list worn outfits newest first with date, occasion, pieces, rating,…	UC-13	US-91	API-OH01	S2 (built early)
FR-46	rate a worn outfit from 1 to 5.	UC-12	US-62	API-O02	S3
FR-47	store wears, swaps, rejections with one of seven reasons, favourites…	UC-09	US-37, US-60, US-61	API-O03, RE-04, RE-16	S2
FR-48	detect patterns from rejections, favourites and wear history and show…	UC-22	US-40, US-63	LR-01 – LR-05, API-LR01	S3 (built early)
FR-49	change the profile only when the user confirms a pattern, shall never…	UC-22	US-40	API-LR01, API-LR02	S3 (built early)
FR-50	use confirmed preferences and repeated rejection reasons when it…	UC-09	US-60, US-63	EX-05, API-EX02, DR-06	S3 (built early)
FR-73	When two or more items deleted as “No longer my style” share a style,…	UC-22	US-34, US-40	DR-04, DR-05, API-DR04	S3 (built early)
FR-51	map each message to exactly one of 14 whitelisted functions, run it…	UC-07	US-31, US-76	AI-01 – AI-16, AI-20 – AI-27, API-AC01 – AC03, AC05 – AC10, DU-03	S3 (built early)
FR-52	list the function it used and offer Undo for every change it made.	UC-07	US-76	API-AC04, API-AC08	S3 (built early)

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-53	answer with the keyword parser when no language model is configured,…	UC-07	US-31	AI-17 – AI-19	S3 (built early)
FR-74	add an item to the shopping list through Ask Closet — “put beige…	UC-07, UC-14	US-36	SL-01, SL-02, API-SL01 – SL04	S3 (built early)
FR-54	utilisation (pieces worn in the last 30 days ÷ pieces owned) and the…	UC-13	US-52, US-69	API-O02, API-O04	S3
FR-55	list forgotten pieces (worn before, not in the last 40 days) with the…	UC-08, UC-13	US-54	API-O04, RE-14	S3
FR-56	compute a Wardrobe Health score from 0 to 100 using the published…	UC-13	US-89	WH-01 – WH-05, API-H01, API-H02	S3 (built early)
FR-57	suggest pieces for real gaps and missing basics, each with a reason,…	UC-14	US-32, US-33	API-SH01, API-AC05	S3 (built early)
FR-72	recommend a replacement item for every item deleted as “Torn /…	UC-05, UC-14	US-33	DR-02, API-DR01, API-DR02, API-SH02	S3 (built early)
FR-58	avoid treating items deleted as “No longer my style” as shopping…	UC-14	US-34	DR-03, API-DR03	S3 (built early)
FR-59	answer “Do I need this?” with one verdict (Good buy, Optional, Skip,…	UC-14	US-88	API-SH04, API-AC03	S3 (built early)
FR-60	warn before a purchase when the user owns pieces similar to the one…	UC-14	US-65	DU-01 – DU-03, API-DU01, API-DU02	S3 (built early)
FR-61	keep a shopping list whose items show their source — Added by Ask…	UC-14	US-36	API-SL04, API-SL05, API-SH03	S3 (built early)
FR-62	notify the user of: an event tomorrow without a planned outfit, a…	UC-21	US-92	API-N01, API-N02, API-K03, API-PL02	S3 (built early)
FR-63	keep read / unread state per notice and highlight at most three…	UC-21	US-92, US-74	API-N01	S3 (built early)
FR-64	propose one outfit per trip day from Available pieces, re-using…	UC-20	US-68	PK-01, PK-03, API-PK01	Stretch (preview)
FR-65	warn about rain without a suitable outer layer, cold without a warm…	UC-20	US-68	PK-02	Stretch (preview)
FR-66	set every listed Available piece to Packed with a return date the day…	UC-20	US-68	API-PK02, API-PK03	Stretch (preview)
FR-67	list accounts (email, role, status, created, piece count) and suspend…	UC-15	US-80	API-AD02 – AD04	S3 (built early)

FR	Requirement (short)	Use case	Stories	Tests	Release
FR-68	refuse every administration function to a non-admin (403).	UC-15	US-80	API-AD01, API-AD05, UT-S08	S3 (built early)
FR-77	give an Admin no route to any user’s pieces, photos, outfits,…	UC-15	US-80	API-AD02	S3 (built early)
FR-69	service health (database, weather, language model, try-on,…	UC-16	US-81, US-83	API-AD05	Stretch (preview)
FR-70	record each successful change of NFR-15 with its time and the piece’s…	UC-01	US-93	API-N02, API-N03	S2 (built early)

6.5 Global, cultural, social, environmental, economic, public-health and welfare considerations

Area	How Closet addresses it	Requirements
Global	Local weather for any city and destination; °C; layout ready for other languages	FR-29, FR-64, NFR-11
Cultural	Modest option respected everywhere; local occasions (weddings, interviews) with default dress codes	FR-07, FR-30, BR-4
Social	Comments only on clothes, never on bodies; size is a private field	NFR-10
Environmental	Utilisation, forgotten pieces, Wardrobe Health and the duplicate warning help people wear what they own and buy less	FR-54, FR-55, FR-56, FR-60
Economic	Shopping suggests only real gaps and replacements; “Do I need this?”; free to use; AI cost capped	FR-57, FR-58, FR-59, NFR-12
Public health	Weather-appropriate outfits; heat and cold notes; less daily decision stress	FR-29
Safety and welfare	Private by default; export and deletion; learned data visible and undoable; at most 3 highlighted notices a day	NFR-05, NFR-13, NFR-14

Appendix A: Glossary

The client, the recommendation engine, the AI chat and the database all use these exact values, so “business casual” means the same thing everywhere.

Dress-code level	1 Casual	2 Smart Casual	3 Business Casual	4 Semi-Formal	5 Formal
Default for occasions	Everyday, University, Travel	Dinner, Date night	Work, Interview	Party	Wedding

Term	Allowed values and rules
Styles	Casual, Chic, Minimal, Elegant, Classic, Sporty, Streetwear, Comfortable, Bold
Availability status	Available, Laundry, Borrowed, Packed, Unavailable – only Available pieces are ever used in an outfit
Rejection reasons (7)	Too formal · Too casual · Don’t like these colours · Not comfortable · Wore this recently · Don’t like one piece · Not in the mood today (hides the look for 7 days, no lasting change)
Pattern answers (after 2 rejections of a piece)	I don’t like it anymore (excluded) · Just not these days (paused 2 weeks) · Keep suggesting it (count reset)
Rating after wearing	1–5 stars or Loved it (raises similar colours and styles) · Wouldn’t repeat (combination retired)
Deletion reasons (~~4~~ 6)	Became too small → same item, new size · Became too big → same item, new size · Torn or damaged → replacement · No longer my style → never suggested again · ~~Donated, sold or other~~ Gave it away · Other → no suggestion
Categories (7)	Top · Dress · Bottom · Layer (shown as Outerwear) · Shoes · Bag · Extra (shown as Accessories)
Outfit	Up to 4 pieces: top or dress, bottom or layer, shoes, ~~extra~~ a bag or an accessory
Match % (version 1.2)	Average of four graded parts, 0 – 100: Colour, Style, Formality (a piece one level away gets half), Weather; shown next to the rule score in Mix & Match and on every recommendation
Recycle time	0 – 30 days (default 3): tops, bottoms and dresses worn more recently are not suggested
Laundry basket time	1 – 14 days (default 2): when a piece in Laundry is expected back; with auto-return it becomes Available then
Match criteria (builder)	Complete outfit · Colour harmony (≤ 2 accent colours, no clash) · Dress code (every main piece suits the level) · Weather fit (≥ 26 °C no warm pieces; < 19 °C a warm layer) · Style consistency (+ Modest when on)
Rule score	100 − 17 per failed criterion; always labelled “rule score”, never a fashion verdict
Utilization	Pieces worn in the last 30 days ÷ pieces owned
Forgotten piece	Worn at least once, but not in the last 40 days
Match % (v1.3 wording)	Average of four part scores (Colour, Style, Formality, Weather), each from a fixed rule: Colour 40 (clash) / 55 (> 2 accents) / 88 (2) / 96 (1) / 92 (all neutral); Formality = share of main pieces at the exact level, half credit one level away; Style = share of main pieces in the chosen styles; Weather 98 / 95 / 45 / max(30, 90 − 30 × warm pieces) by temperature. Deterministic, not a trained model.
Wardrobe Health	0 – 100 = Utilisation 35 × (worn in 30 days ÷ owned) + Versatility 20 × (pieces with ≥ 2 dress codes ÷ owned) + Balance 20 × (1 − gaps ÷ (3 × levels dressed for)) + Readiness 15 × (Available ÷ owned) + Basics 10 × (basics owned ÷ 7) − 1 per forgotten piece (max 10) − 1 per potential duplicate (max 10). Grades: Excellent ≥ 80, Good ≥ 65, Fair ≥ 50, Needs care.
Potential duplicate	A piece with the same category and colour as another piece and at least one shared style
Similarity (shopping)	Same category required; colour 35 (same) or 15 (both neutral) + category 25 + style 15 × shared ÷ all + formality 15 × shared levels ÷ all + season 10; ≥ 70 = similar

Term	Allowed values and rules
Learned pattern · confidence	A rule over the user's own feedback (e.g. ≥ 2 "Too formal" rejections in 60 days). Confidence = evidence ÷ (evidence + 2).
Conflict	A planned piece that is not Available and will not be back by the event date, or one piece planned for two events < 4 h apart
Occasion template	A preset of occasion, dress-code level, style and weather sensitivity (outdoor / indoor)
Activity log	The user's own list of successful changes (NFR-15)
Built early / preview	A working feature of a later sprint, labelled in the app, accepted only at its own sprint review

Acronyms. API — application programming interface · BR — business rule · FR / NFR — functional / non-functional requirement · JWT — JSON web token · LLM — large language model · MoSCoW — Must, Should, Could, Won't · PDPL — Egypt's Personal Data Protection Law · SDD — software design description · SRS — software requirements specification · UC — use case · US — user story · WCAG — Web Content Accessibility Guidelines.

Appendix B: Analysis Models

B.1 Use case model. Two actors — the User and the Admin — and six external services (Figure B-1). Colour shows the sprint that delivers each use case. Figure B-2 zooms into the two use cases with the most structure.

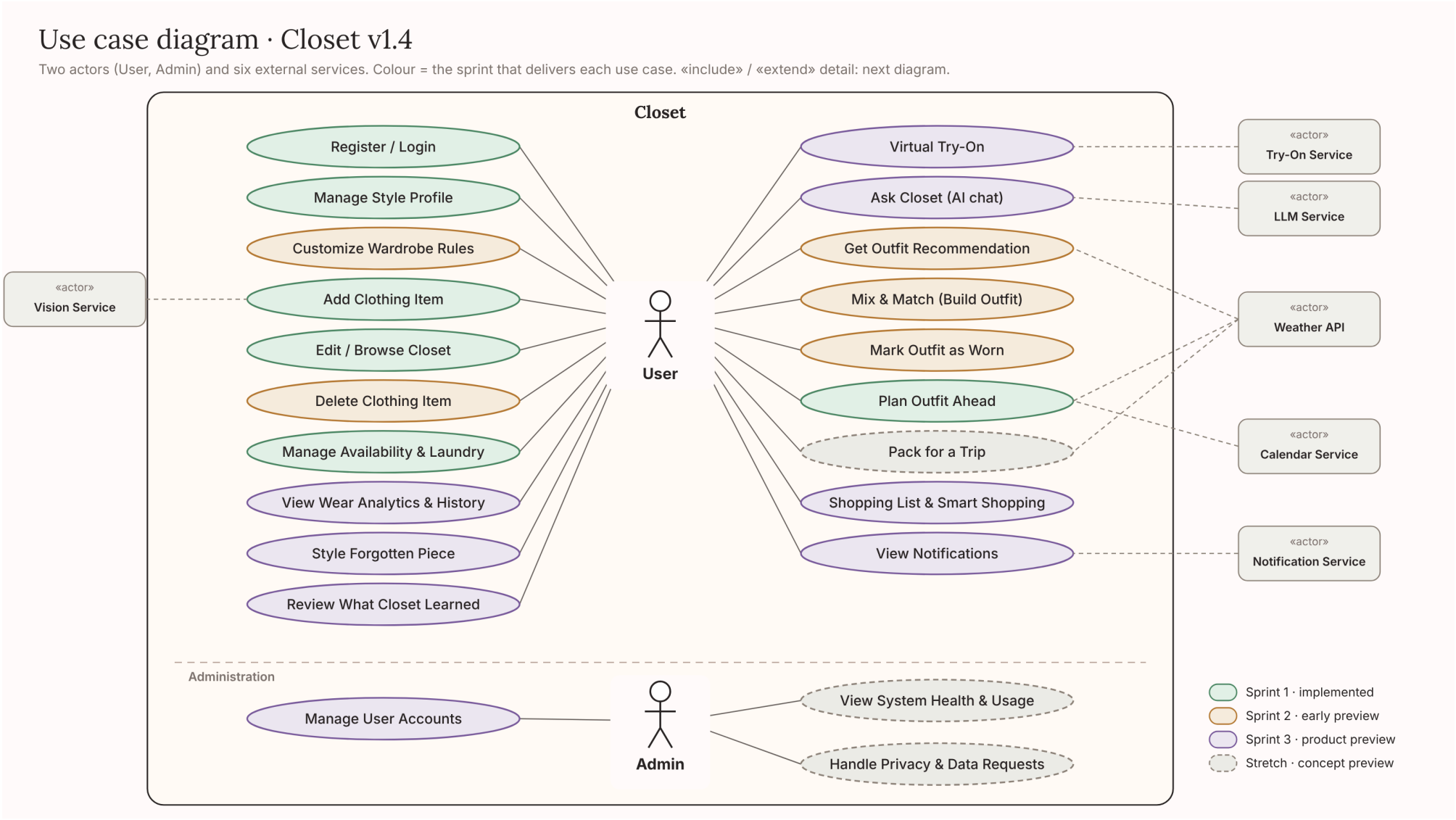


Figure B-1 – Use case diagram, version ~~1.3~~ 1.4 (prepared by Bashar). Associations connect each actor to the use cases it starts or takes part in; «include» and «extend» relationships of UC-09 and UC-07 are shown in Figure B-2.

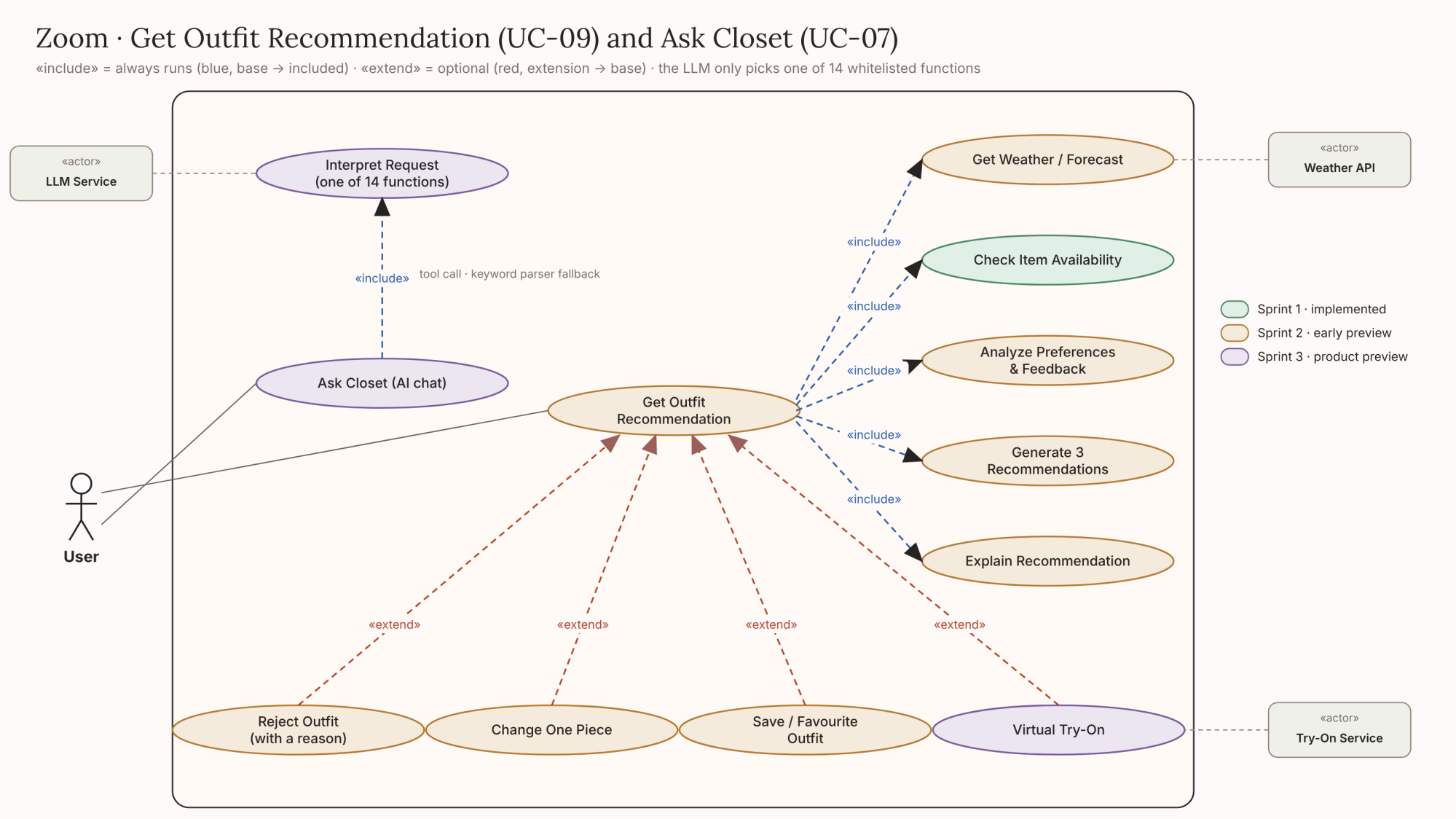


Figure B-2 – Get Outfit Recommendation and Ask Closet in detail. «include» runs every time; «extend» is optional behaviour.

Actor	Type	Role
User	Primary	Owns a closet; starts every user use case
Admin	Primary	Manages accounts and service health; never sees wardrobe contents
Vision Service	External	Suggests a piece's tags from its photo (stretch)
LLM Service	External	Interprets a chat message into one of ~~13~~ 14 whitelisted functions; never sees photos or the database
Virtual Try-On Service	External	Photo-real try-on (future); today a preview on the user's own photo drawn in the browser, with the stylised avatar as fallback
Notification Service	External	Push / e-mail (future); in-app today
Weather API (Open-Meteo)	External	Forecast for the user's city or a trip destination
Calendar Service	External	Events from an exported calendar file

B.2 Use case descriptions.

Use case	Main flow	Relationships	Alternatives and exceptions	Stories
UC-01 Register / Login	Enter email + password or Google → account opened; first login starts the quiz; Sign out ends the session	— «include» Verify Credentials; «extend» Sign in with Google. Delete Account «include» Confirm with Password	Wrong password → error; 5 failures lock 15 min; suspended account → refused; delete account from Settings (double confirmation)	US-01, US-06
UC-02 Manage Style Profile	Edit styles, colours, occasions, dress code, modest, city~~; view and undo lessons~~	«extend» Complete Style Quiz (first login)	Quiz skipped → defaults	US-02–05~~, US-40~~
UC-03 Add Clothing Item	Take or pick photo → enter details → review → Save → piece Available	«include» Upload Photo; «extend» AI Pre-fill Details	Vision down or > 10 s → manual entry; missing field → Save disabled	US-08, 09, 11
UC-04 Edit Clothing Item	Browse and filter → open piece → edit details	—	From the piece: Style this piece, change status, delete	US-12, US-71
UC-05 Delete Clothing Item	Remove → choose 1 of 4 reasons (Torn / damaged, No longer my style, Became too small, Other) → the effect is shown → piece removed, reason kept	«include» Record Deletion Reason; «include» Apply Deletion Effect (BR-10)	Piece in a saved or planned outfit → user is warned, then it is removed from those outfits; Torn / damaged or Became too small → a replacement appears in the shopping list; No longer my style → never suggested again, may become a learned pattern (UC-22)	US-13, US-33, US-34
UC-06 Manage Availability & Laundry	Set one of 5 statuses; laundry basket lists Laundry pieces with the day each is back	«extend» Mark All as Clean; «extend» Set Return Date & Note	— Laundry basket time passed and auto-return on → piece becomes Available by itself; return date in the past → refused	US-15, US-16, US-56, US-84, US-86

Use case	Main flow	Relationships	Alternatives and exceptions	Stories
UC-07 ~~Chat with AI Closet Agent~~ Ask Closet (AI chat)	Message → LLM picks a whitelisted function → run on the user's data → reply lists the function, offers Undo	«include» Interpret Request~~;~~ ~~«include» Analyze Prefs & Feedback~~; «extend» Undo Agent Action	No match → one clarifying question; LLM down or > 8 s → keyword parser; "put beige trousers on my shopping list" / "add this to my shopping list" → item added to the shopping list (UC-14), Undo offered	US-31, US-76, US-36
UC-08 Style ~~This /~~ Forgotten Piece	Pick a piece → 3 outfits that all contain it	«include» ~~UC-09~~ Generate 3 Recommendations	Piece not Available → offer to change status	US-59
UC-09 Get Outfit Recommendation	Detailed below; an occasion template can fill the choices (US-90)	~~4~~ 5 «include», 4 «extend»	Detailed below	US-17, 19–21, 24, 26, 27, 37, 38, 39, 51, 57, 60, 61
UC-10 Plan Outfit Ahead	Add or pick an event (manual, or synced as a stretch goal) → 3 outfits for its dress code and forecast → choose one	«include» Read Calendar Event; «include» ~~UC-09~~ Get Weather / Forecast; «include» Generate 3 Recommendations; «extend» Receive Outfit Reminder; Read Calendar Event «extend» Import Calendar (.ics)	No calendar connected → manual events; a piece in the laundry or borrowed counts if it is back by the event; a planned piece that will not be back → conflict with a one-tap replacement (US-79)	US-46–48, US-87
UC-11 Build Outfit	Pick 4 pieces → pass/fail per criterion, rule score, one-sentence mismatch	«include» Check Outfit Compatibility	Suggested swap (stretch)	US-42, US-43, US-66
UC-12 Mark Outfit as Worn	Wear this → wear count and last-worn updated for all 4 pieces	«include» Update Wear History; «include» Update Wear Analytics; «extend» Rate Outfit After Wearing	Undo in the same session	US-23, US-62
UC-13 View Wear Analytics	Utilization score for the last 30 days; Wardrobe Health with its parts; outfit history with filters	«extend» Most Worn, Least Worn, Forgotten Pieces, Utilization Score	No history yet → explanation	US-52, US-54, US-69, US-89, US-91
UC-14 View Smart Shopping List	Deletion reasons → replacements; gaps by category; duplicates flagged; list shown with a reason per item; each item tagged with its source (Ask Closet, Replacement, Wardrobe gap, Smart suggestion); mark bought or remove	«include» Analyze Wardrobe Gaps; «include» Check Similar Owned Items; «include» Read Deletion Reasons; «extend» Check Before Buying	Nothing needed → says so	US-32–34, US-36, US-64, US-65 (stretch), US-88
UC-15 Manage User Accounts	Admin searches by email → sees status, creation date, last sign-in, number of pieces	«include» Verify Admin Role; «extend» Suspend / Reactivate Account	Not an admin → 403, attempt logged	US-80
UC-16 View ~~System Health &~~ Usage Statistics	Uptime, error rate, adapter fallback counts, totals of users and pieces	«include» Verify Admin Role	Totals only, no individual data	US-81

Use case	Main flow	Relationships	Alternatives and exceptions	Stories
UC-17 Handle Privacy & Data Requests	Export or delete a user's data on request; runs the same deletion as US-06	«include» Verify Admin Role	Every request written to the audit log	US-82
~~UC-18 Manage App Configuration~~	Edit category, dress-code and occasion lists; feature switches	«include» Verify Admin Role	Switch the AI chat off when the language model is down	US-83
UC-19 Customize Wardrobe Rules	Settings → set laundry basket time (1 – 14 days), auto-return, laundry day, recycle time (0 – 30 days), reminders and their hour	—	Out-of-range value → field error; recycle time 0 → no limit	US-84, US-85
UC-20 Pack for a Trip	Destination, dates, trip type and activities → a suitcase of Available pieces with one outfit per day → Add to packing list → Mark as packed	«include» Get Weather / Forecast; «include» Generate 3 Recommendations (per day)	No outfit for a day → reported as missing; a piece no longer Available → skipped and reported; no live forecast → seasonal estimate with a warning	US-68
UC-21 View Notifications	Open the bell → notices from planner, availability, packing, analytics and weather, newest and most important first → open one → it is marked read	—	Nothing to show → "You're all caught up"; at most 3 highlighted a day	US-92
UC-22 Review What Closet Learned	See each detected pattern with evidence and confidence → Confirm (profile changes) or Not accurate (dismissed) → Undo restores the previous values	—	Too little evidence → no pattern shown; unknown pattern → refused	US-40, US-63

UC-09 Get Outfit Recommendation (detailed)

Field	Description
Actors	User (primary), Weather ~~Service~~ API; Virtual Try-On Service (extension)
Precondition	Logged in; at least one Available top or dress, bottom and shoes
Main flow	1) User picks occasion and style (dress code, modest, repeats under More options) and a day. 2) «include» Get Weather / Forecast (Open-Meteo). 3) «include» Check Item Availability: only Available pieces that are not disliked, paused or hidden, and not worn within the recycle time. 4) «include» Analyze ~~Prefs & Feedback~~ User Preferences. 5) «include» Generate ~~& Explain~~ 3 ~~Outfits~~ Recommendations: 3 different outfits, each with a rule score, a match % (colour, style, formality, weather) and 3–5 reasons. 6) «include» Explain Recommendation.
Extensions	Virtual Try-On (avatar), opened from any of the three outfits, moving between the three, returning to the same outfits · Save Outfit, extended by Favorite Outfit · Change One Piece (only that slot changes; the changed outfit can then be tried on) · Reject Outfit, extended by Give Rejection Reason and Temporarily Hide Item · Wear this → Rate (1 – 5)
Alternatives	Too few pieces for the dress code → relax one slot and say so · Weather down → last known or manual temperature · Same piece rejected twice → pattern question · Wear this → UC-12
Postcondition	3 explained outfits shown; every reaction stored as feedback

B.2b Deleting a piece: the reason and its effects. Figure B-4 is the analysis model behind BR-10, FR-15 and FR-71 – FR-73.

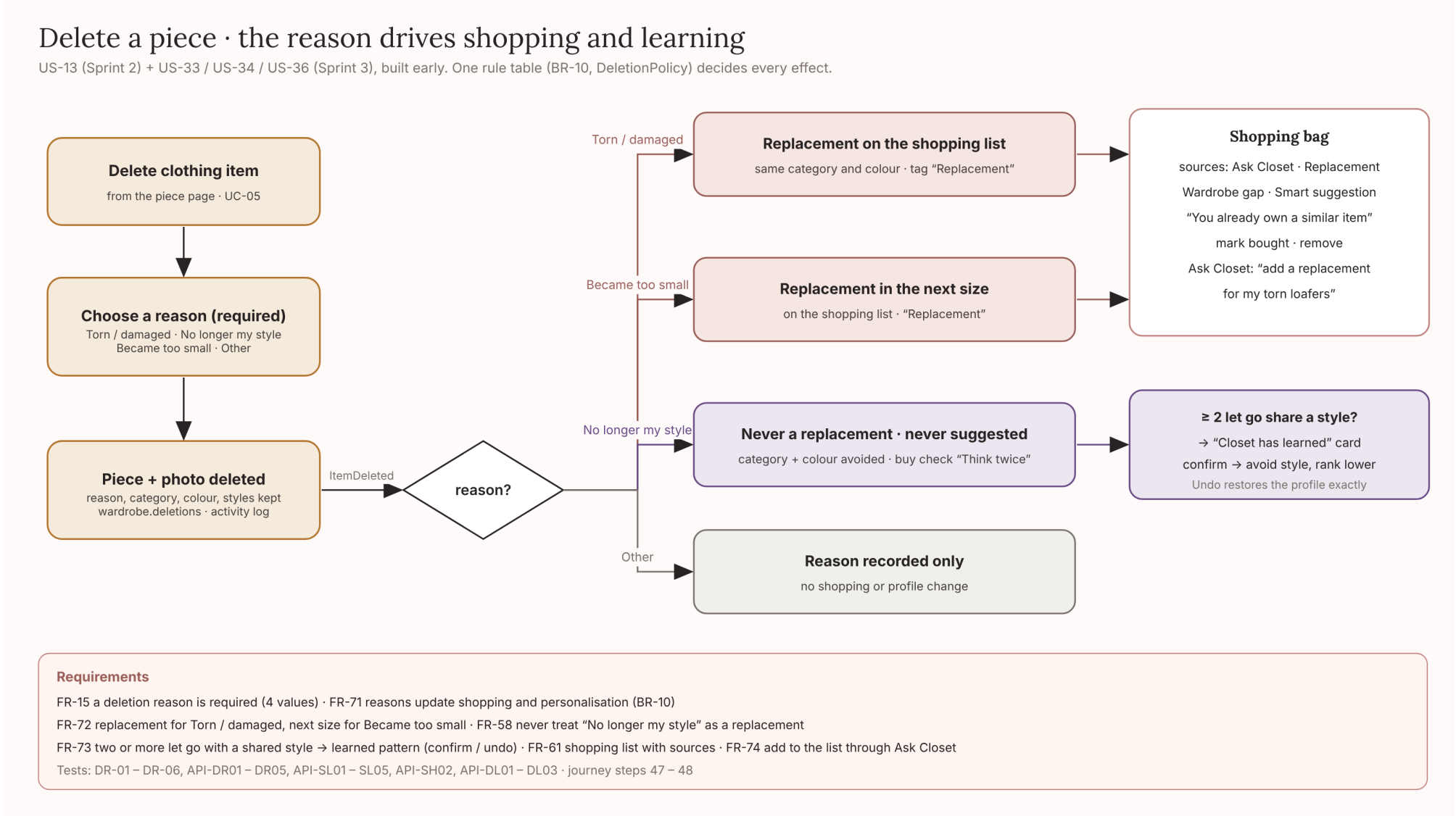


Figure B-4 – Delete a piece with a reason: effects on the shopping list and on learning.

B.3 State model of a piece. The status of a piece is the central domain state; Figure B-3 is the analysis model that BR-1 and BR-2 refer to.

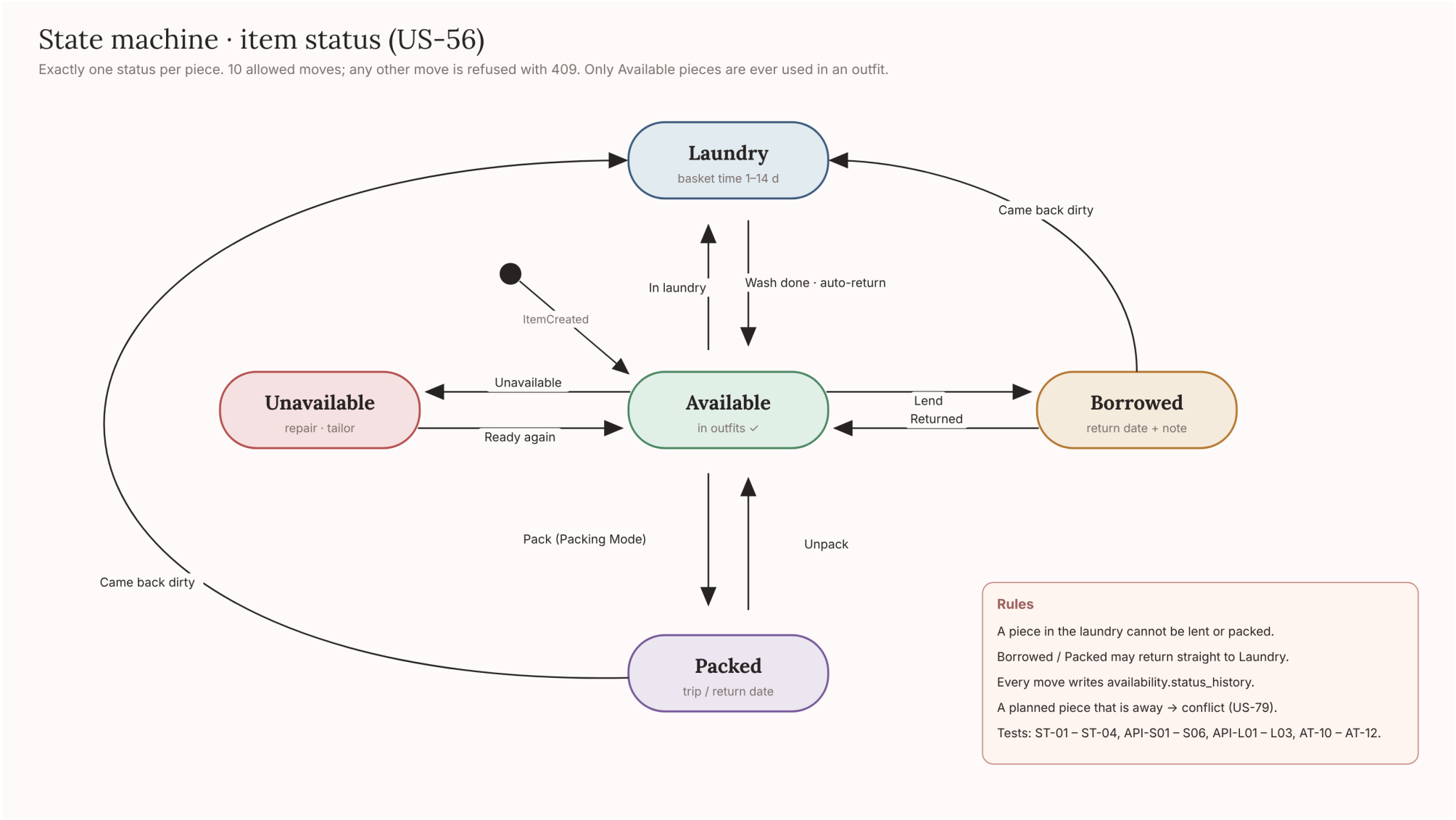


Figure B-3 – Item status state machine.

Appendix C: To Be Determined List

ID	Item	Owner	Due
TBD-1	Load test with a 300-piece closet to confirm NFR-01 (only functional tests so far)	Youssef	Release week
TBD-2	Usability test with 5 users for NFR-03	Habiba	Sprint 3
TBD-3	Per-user time zone for calendar import (Africa/Cairo is assumed)	Bashar	Sprint 3
TBD-4	Which notification service and channel (push or e-mail) for US-67	Karim	Stretch
TBD-5	Vision provider and confidence threshold for US-10	Bashar	Stretch
TBD-6	Accessibility audit tool and the full WCAG 2.1 AA check for NFR-07	Salma	Release week

Appendix D: Change Log

Changes C-01 – C-16 were made for Milestone 2 (Sprint 1 review), C-17 – C-27 after the second design review, C-28 – C-40 for v1.3 (product requirements and the split into SRS and SDD), C-50 – C-57 in revision 2 (deletion reasons, shopping list, core flow) and C-64 – C-67 in revision 3 (final pass: photo try-on, look names, change-piece tray, forgotten piece, Suggest a fix). Design changes C-41 – C-49, C-58 – C-63 and C-68 – C-71 are in the SDD, Appendix E. Section numbers refer to the combined specification where the change was first made.

ID	Type	Section	Version 1.0	Version 1.1	Reason	Stories
C-01	Requirement change	3.3, 3.4, 8	One primary actor; 14 use cases	Admin actor; UC-15 Manage User Accounts (extended by Suspend / Reactivate Account), UC-16 View System Health & Usage, UC-17 Handle Privacy & Data Requests, UC-18 Manage App Configuration, all including Verify Admin Role	Milestone 1 instructor feedback	US-80 – US-83
C-02	Design change	4.2 – 4.4	No authentication or security components	API-tier pipeline: API Gateway (HTTPS, CORS, security headers, rate limits), Request Validation, Authentication (JWT, IAuthToken), Access Control (role + ownership); Identity & Auth module; Secrets & Config store; Admin module and Admin Console (Sprint 3)	Milestone 1 instructor feedback	US-01, US-80
C-03	Requirement change	3.5	NFR-04 listed 5 short rules	NFR-04 lists every security rule with a test: HTTPS, bcrypt, 2-hour tokens ended by logout, lockout, role and ownership checks, headers, CORS, rate limits, validation, safe errors, secrets in environment variables	Milestone 1 feedback; Sprint 1 implementation	US-01
C-04	Requirement change (scope)	3.3, 5	US-61 in Sprint 1	US-61 in Sprint 2	It hides a piece from recommendations, which are built in Sprint 2	US-61
C-05	Requirement change (scope)	3.2, 3.3, 5	US-65 in Sprint 3	US-65 is a stretch story; US-80 takes its 3 hours in Sprint 3	Keeps Sprint 3 at 78 h	US-65, US-80
C-06	Requirement change (refinement)	3.3	US-01 criteria did not mention logout or lockout	Logout ends the session; 5 failed sign-ins lock the account for 15 minutes	Was only in the UC-01 alternatives; now testable	US-01
C-07	Requirement change (refinement)	3.3	US-02, US-03, US-04 criteria depend on recommendations and weather	Each criterion split into a Sprint 1 part (data saved) and a part verified in Sprint 2 with US-19, US-21, US-20	Criteria named features from a later sprint	US-02 – US-04
C-08	Requirement change (refinement)	3.3	US-08: JPG/PNG/HEIC ≤ 10 MB	Adds WebP; the type is checked from the file content; location data is removed; exactly 10 MB is accepted	Sprint 1 implementation; defect D-02	US-08

ID	Type	Section	Version 1.0	Version 1.1	Reason	Stories
C-09	Design change	4.5, 4.9, 6	Object storage “private, encrypted”	Private storage behind a PhotoStorage interface (local disk in Sprint 1, cloud bucket later); HMAC-signed URLs valid 15 minutes; EXIF location removed; encryption at rest provided by the host	What Sprint 1 actually stores	US-08, NFR-05
C-10	Design change	4.3, 8	Closet gaps under Ask Closet / Analytics (UC-14)	Closet gaps computed by Wardrobe, which owns the data	US-64 is in Sprint 1; Ask Closet and Analytics come in Sprint 3	US-64
C-11	Design change	4.4	3 events, 8 interfaces	Adds events UserRegistered and ItemCreated and interfaces IAuthToken and IAdminAPI	Sprint 1 implementation	US-01, US-08
C-12	Design change	4.11 (new)	High-level design only	Lower-level design: ER diagram, class diagram, two sequence diagrams, status state diagram, API table	Milestone 2	Sprint 1
C-13	Design change	3.1, 4.11	Statuses listed, transitions undefined	10 allowed status transitions; any other move is refused	Needed to implement and test US-56	US-56, US-15, US-16
C-14	Implementation detail	4.10	“Node.js + Express”, “React”	Node 22, Express 4, pg, bcryptjs, jsonwebtoken, helmet, express-rate-limit, multer; React 19 + Vite; Jest + Supertest; GitHub Actions; Render blueprint	Sprint 1 implementation	—
C-15	Requirement change (backlog)	3.2	70 stories (43 / 11 / 16; 50 / 15 / 5)	74 stories (43 Must / 12 Should / 19 Could; 50 committed / 16 stretch / 8 future)	C-01 and C-05	US-65, US-80 – US-83
C-16	Design change	Figures 1 and 2	Diagrams version 1.0	Use case diagram and component diagram version 1.1	C-01, C-02	—
C-17	Requirement change (new stories)	3.2, 3.3	74 stories	79 stories: US-84 laundry basket time, US-85 recycle time, US-86 return date and note, US-87 calendar import and wardrobe dates, US-88 “Do I need this?” check	Review comment: customization, calendar	US-84 – US-88
C-18	Requirement change (new use case)	3.4	18 use cases	UC-19 Customize Wardrobe Rules	Review comment: customization	US-84, US-85

ID	Type	Section	Version 1.0	Version 1.1	Reason	Stories
C-19	Requirement change (refinement)	3.4	UC-09 included 4 steps	UC-09 «include» Get Weather, Check Item Availability, Analyze User Preferences, Generate 3 Recommendations, Explain Recommendation; «extend» Change One Piece, Virtual Try-On, Reject Outfit, Save Outfit. Style Forgotten Piece and Plan Outfit Ahead «include» Generate 3 Recommendations; Plan Outfit Ahead also «include» Read Calendar Event and Get Weather / Forecast. Mark Outfit as Worn «include» Update Wear History and Update Wear Analytics	Full-product model; every included step is a real function in the engine	US-19 – US-24, US-47, US-59
C-20	Requirement change (actors)	3.4	4 external services	6: Vision, LLM, Virtual Try-On, Weather API (Open-Meteo), Calendar, Notification	Review comment: role of the LLM; full system	—
C-21	Requirement change (authentication)	3.4	Authentication inside UC-01 text only	Verify Credentials, Sign in with Google, Confirm with Password drawn as use cases	Review comment: authentication not evident	US-01, US-06
C-22	Design change	4.2 – 4.4	Component diagram v1.1	v1.2: Smart Shopping, Reminders, Try-On components; Try-On, Calendar and Notification adapters with their external services; notes on the LLM’s role and on Availability; authentication note in the API tier	Review comments	—
C-23	Design change (data)	4.5, 4.11	Schema 1.1	Migration 003: Bag category and piece names; profile settings (display name, laundry days, auto-return, laundry weekday, repeat days, reminders); status return date and note; wardrobe.deletions; shopping.decisions	C-17	US-13, US-84 – US-88
C-24	Requirement change (vocabulary)	3.1	6 categories	7 categories: Bag added (shown as Bags); an outfit’s fourth piece is a bag or an accessory; 6 deletion reasons	Demo wardrobe and recommendations needed bags as a slot	US-09, US-13
C-25	Design change (Ask Closet)	4.3, 4.7	Ask Closet owned the smart shopping list	Smart Shopping is its own component (S3); Ask Closet calls it for “what is missing” and “do I need this”. 13 whitelisted functions	One responsibility per component	US-32 – US-36, US-88
C-26	Implementation status	5	Sprint 2 and 3 not started	Early previews of Sprint 2 and Sprint 3 features run on real data and are labelled in the app; none is counted as Sprint 1 work	Show the full product honestly	—
C-27	Design change	Figures 1 and 2	Diagrams v1.1	Diagrams v1.2; the use case diagram colours each use case by sprint	C-18 – C-22	—

ID	Type	Section	Version 1.0	Version 1.1	Reason	Stories
C-28	Requirement change (scope)	—	US-79 Stretch	US-79 conflict detection in Sprint 2 (built early) with double-booking	Strongest demonstration of modules working together; asked for in the final review brief	US-79
C-29	Requirement change (scope)	—	US-65 Stretch	US-65 duplicate-purchase warning in Sprint 3 with a defined similarity score	Most useful shopping use case; similarity made measurable	US-65
C-30	Requirement change (scope)	—	US-68 Future	US-68 packing mode: Stretch, working concept preview (UC-20)	Uses the existing Packed status; shows Weather + Engine + Planner + Availability together	US-68
C-31	Requirement change (scope)	—	US-81 Future	US-81 admin system health: Stretch, working preview	Instructor feedback on the Admin actor; a coherent Admin experience	US-81
C-32	Requirement change (refinement)	—	US-57: 3 – 5 reasons	US-57: “Recommended because …” sentence + factors from facts the engine used + defined match % (FR explain, breakdown)	Explainable AI; no meaningless percentages	US-57
C-33	Requirement change (refinement)	—	US-40 in UC-02	US-40 is its own use case UC-22 with evidence, confidence, Confirm / Not accurate and exact Undo	Make the learning loop visible	US-40, US-63
C-34	Requirement change (new stories)	—	79 stories	84 stories: US-89 Wardrobe Health, US-90 occasion templates, US-91 outfit history, US-92 Notification Center, US-93 activity log	v1.3 product requirements	US-89 – US-93
C-35	Requirement change (new use cases)	—	19 use cases	UC-20 Pack for a Trip, UC-21 View Notifications, UC-22 Review What Closet Learned	C-30, C-33, C-34	US-68, US-92, US-40
C-36	Requirement change (NFR)	—	13 NFRs	NFR-14 safety (no unintended change, exact Undo), NFR-15 auditability; NFR-07 accessibility made specific (keyboard, alt text, reduced motion)	Activity log, learning and accessibility polish	US-93
C-37	Requirement change (business rules)	—	Rules in text	Business rules BR-1 – BR-9 listed in 5.5, including BR-9 (every score from a defined rule)	Template section 5.5; honest scoring	—
C-38	Document change	—	One combined specification	Separate SRS (this document, IEEE / Wiegers template) and SDD (IEEE 1016 template)	Requested structure for Milestone 2	—
C-39	Requirement change (IDs)	—	Requirements only as user stories	FR-01 – FR-70 added per system feature; US and UC IDs unchanged	Template section 4; traceability	all

ID	Type	Section	Version 1.0	Version 1.1	Reason	Stories
C-40	Requirement change (UI)	—	Generic screens	UI-1 – UI-5: sprint labels on every screen, one goal and one physical metaphor per screen, empty / loading / success states, labelled scores	Final review brief; usability	—
C-50	Requirement change (core flow)	—	Outfits listed among other features	Section 2.1 states the core loop first (three looks → change a piece → try on → feedback → learning → shopping), Figure 2-2; FR-75 presentation and actions of the three outfits; FR-76 try-on flow	Team review: the heart of the product was under-stated	US-19, US-24, US-27, US-26, US-23, US-60, US-62
C-51	Requirement change (refinement)	—	FR-15: one of six reasons	FR-15: a deletion reason is required — exactly four: Torn / damaged, No longer my style, Became too small, Other (the M1 backlog already said “4 reasons”)	Each reason has one defined effect	US-13
C-52	Requirement change (new)	—	—	FR-71 deletion reasons update shopping and personalisation; BR-10 the effect of each reason	Deletion reasons are product logic, not a UI detail	US-13, US-33, US-34
C-53	Requirement change (new / refinement)	—	FR-57 mixed replacements with gaps; FR-58 avoid only	FR-72 replacement for Torn / damaged and next size for Became too small, on the shopping list; FR-57 now gaps and basics; FR-58 “No longer my style” never treated as a replacement, “Think twice” on a buy check	Make the shopping logic explicit and testable	US-32, US-33, US-34
C-54	Requirement change (new)	—	—	FR-73 learning from “No longer my style” deletions (two or more with a shared style → learned pattern; confirm / Undo)	Close the loop from deletions to future recommendations	US-34, US-40
C-55	Requirement change (refinement / new)	—	FR-61: add, remove, bought	FR-61 shopping list with sources and similar-item warning (BR-11); FR-74 add to the list through Ask Closet; FR-51 14 functions	Shopping list as a real product section	US-36
C-56	Requirement change (new)	—	BR-3 only	FR-77 Admin has no route to wardrobe contents	Clean Admin role (instructor feedback)	US-80
C-57	Requirement change (release label)	—	Try-on shown as a Sprint 2 preview in some screens	Virtual try-on is US-27, Sprint 3 (built early) everywhere: app, diagrams D1/D2, SRS	Consistency between app, backlog and documents	US-27
C-64	Requirement change (new)	—	—	FR-78 each of the three outfits has its own name, never two alike	Two looks were both called “Easy Striped Top” in the demo	US-19
C-65	Requirement change (refinement)	—	FR-34 best alternative swapped in silently	FR-34 up to three Available alternatives, each with its recalculated match %; the user picks one	Make “change one piece” visible and controllable	US-24

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C-66	Requirement change (refinement / new)	—	FR-35 three outfits around a piece	FR-35 every outfit contains the chosen (e.g. forgotten) piece, which must be Available; weather and recycle time do not remove it; FR-79 Suggest a fix in Mix & Match with the score before and after	Defect D-06; showcase forgotten-piece styling and Mix & Match	US-54, US-59, US-66
C-67	Requirement change (refinement / new)	—	FR-42 stylised avatar only, no photo of the user	FR-42 Virtual Try-On Preview on the user’s own full-body photo, avatar as fallback, graceful when no provider; FR-80 the photo stays on the device, is never uploaded and can be removed; NFR-05 updated	Instructor feedback: try-on must show the user, with privacy by design	US-27